COHERENT QUADRANT

PEM & AEM Electrolyzers for Green Hydrogen Projects Market

India

30 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

PEM & AEM Electrolyzers for Green Hydrogen Projects Market

Market Type

The PEM & AEM Electrolyzers for Green Hydrogen Projects Market refers to the sales of Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) electrolyzers that use renewable electricity to split water into hydrogen and oxygen, producing green hydrogen. The market covers electrolyzer stacks and integrated systems deployed across industrial and commercial green hydrogen projects. By product: PEM electrolyzers, AEM electrolyzers. By configuration: electrolyzer stacks, integrated electrolyzer systems. By system capacity: below 1 MW, 1 MW to below 5 MW, 5 MW to below 20 MW, 20 MW and above. By application: green ammonia production, petroleum refining, iron & steel manufacturing, chemical manufacturing, energy storage & power, other industrial applications. By end user: green hydrogen project developers, industrial hydrogen producers, captive industrial users, utilities & energy companies. Geography: India -- companies HEADQUARTERED IN India.

Provider Categories in this Market

1. Manufacturer

Set by the market brief (_pipeline/market_brief.json).

30	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

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Market Scoring Parameters

X — Product Capability

1. **PEM/AEM Stack Technology Readiness**
2. **Electrolyzer System Efficiency & Performance**
3. **Manufacturing Capacity & Scalability**
4. **Product Portfolio Breadth & Modularity**
5. **R&D and Innovation Capability**

Y — Business Capability

1. **Domestic Customer Contracts & Order Book**
2. **Revenue & Funding Strength**
3. **Strategic Partnerships & Alliances**
4. **Geographic Footprint & Sales Network**
5. **Market Share & Installed Base**

Quadrant Positioning

(20 on chart · 30 in total)



Countries on chart (by quadrant)

Leaders: India×5 · **Challengers**: India×5 · **Trailblazers**: India×5 · **Evolving Players**: India×5

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Ohmium	Ohmium Operations Private Limited	Bengaluru, India	Manufacturer	Leaders	100	100	100
GreenH	GreenH Electrolysis Pvt. Ltd.	New Delhi, India	Manufacturer	Leaders	92	94	93
HYDGEN	Hydrogen Innovation Private Limited	Mangaluru, India	Manufacturer	Leaders	94	90	92
Shakti Photon Solutions	Shakti Photon Solutions	Chennai, India	Manufacturer	Leaders	91	88	90
Adani	Adani New Industries Limited	Ahmedabad, India	Manufacturer	Leaders	89	92	90
Hydrogenergy	Hydrogenergy Technologies Pvt. Ltd.	New Delhi, India	Manufacturer	Challengers	78	86	82
SSB Green Enviro	S.S.B Green Enviro Pvt. Ltd.	Ahmedabad, India	Manufacturer	Challengers	84	80	82
Sentient Engines	Sentient Engines Pvt Ltd	Nashik, India	Manufacturer	Challengers	77	75	76
Patel Gas Plants	Patel Gas Plants	Vapi, India	Manufacturer	Challengers	77	76	76
Newage Green Electro	Newage Green Electro Private Limited	Hyderabad, India	Manufacturer	Challengers	73	75	74
BHEL	Bharat Heavy Electricals Limited	New Delhi, India	Manufacturer	Trailblazers	84	85	84
homiHydrogen	homiHydrogen Private Limited	Pune, India	Manufacturer	Trailblazers	82	85	84
H2GenX	Airox Nigen Private Limited	Ahmedabad, India	Manufacturer	Trailblazers	87	78	82
YAAI Green Energy	YAAI Green Energy Pvt. Ltd.	Gurugram, India	Manufacturer	Trailblazers	86	75	80
Hypower	Hypower Green Energy Technologies Pvt Ltd	Bangalore, India	Manufacturer	Trailblazers	85	72	78
Gastek Engineering	Gastek Engineering Pvt. Ltd.	New Delhi, India	Manufacturer	Evolving Players	82	73	78
Nanosol Energy	Nanosol Energy Private Limited	Hyderabad, India	Manufacturer	Evolving Players	79	73	76
CLIMAFOUR	CLIMAFOUR Labs Pvt. Ltd.	Jaipur, India	Manufacturer	Evolving Players	80	67	74
Alphios Finergy	Alphios Finergy Private Limited	Chennai, India	Manufacturer	Evolving Players	72	68	70
Hydrovolt Energy	Hydrovolt Energy Ptx Private Limited	Ghaziabad, India	Manufacturer	Evolving Players	72	67	70

Other Noticeable Players

(10 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
Agastyah2	Agastya Eco Ltd	Ahmedabad, India	Manufacturer	Leaders	
Reliance Industries Limited	Reliance Electrolyser Mfg. Ltd.	Mumbai, India	Manufacturer	Leaders	
H2next	H2next Private Limited	Karaikkudi, India	Manufacturer	Leaders	
Hyetron Energy	Hyetron Energy Pvt. Ltd.	Chennai, India	Manufacturer	Leaders	
ORAIPL	Ozone Research & Applications (India) Pvt. Ltd.	Nagpur, India	Manufacturer	Challengers	
Madhuchitt Industries	Madhuchitt Industries LLP	Mumbai, India	Manufacturer	Trailblazers	
LiveHy Energy	Livehy Energy Private Limited	Bangalore, India	Manufacturer	Trailblazers	
PRANAS	VoltHyn Private Limited	Kallakurichi, India	Manufacturer	Evolving Players	
Hynera Energy	Hynera Energy Private Limited	Hyderabad, India	Manufacturer	Evolving Players	
Kitech Hydrogen Systems	Kitech Hydrogen Systems (India) Private Limited	Pune, India	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Ohmium Operations Private Limited

Ohmium · Leaders · X 100 · Y 100 · Overall 100

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 100/100

> The rating rests on the company's verified deployment of its proprietary PEM technology across active commercial pipelines and strategic partnerships, alongside attaining critical global safety and engineering certifications such as CE Mark and UL certifications. Based on primary research, the company's total operational hours logged can be estimated as extensive across major deployments, based on its field records documenting thousands of operational hours validating dynamic ramping capabilities.

SUPPORTING POINTS

- The company reports that its Lotus Proton Exchange Membrane (PEM) electrolyzers are modular, interlocking systems with a dynamic ramping range of 8 seconds from minimum to full load and an industry-leading stack efficiency of 45 kWh/kg.
- Ohmium received CE Mark approval for its Lotus PEM Electrolyzers and achieved a third-generation Lotus footprint of 29.7 m²/MW alongside securing an industrial hydrogen generator certification from UL Solutions.

Electrolyzer System Efficiency & Performance — scored 100/100

> The rating rests on the company's publicly disclosed beginning-of-life electrical efficiency metrics for its commercial-scale PEM electrolyzer systems. Based on primary research, the company's long term operational degradation can be estimated as minimal under real operating conditions, based on its focus on continuous engineering improvements and hypermodular designs.

SUPPORTING POINTS

- The company reports a system efficiency of 48 kWhAC/kg at Beginning of Life (BOL) for its Lotus PEM Electrolyzer Series E (with stack and balance of plant consuming 45 kWh/kg), along with an 8-second dynamic ramping capability.
- The company reports a system efficiency of 49 kWh/kg at Beginning of Life (BOL) for its E series and 53 kWh/kg for its P series.

Manufacturing Capacity & Scalability — scored 100/100

> The rating reflects the company's clear and verifiable gigawatt-scale production capability established via a newly commissioned 2 GW manufacturing plant near Bengaluru that is designed for modular mass production. Based on primary research, the company's manufacturing throughput capability can be estimated as substantial and rapidly scalable, based on its 2 gigawatt doddaballapura gigafactory designed for mass production of fully assembled and tested modular electrolyzer systems.

SUPPORTING POINTS

- Ohmium Operations Private Limited launched its second green hydrogen electrolyzer gigafactory in Doddaballapura, Bengaluru, in July 2024, ramping up to ship 2 GW of fully assembled and tested Proton Exchange Membrane (PEM) electrolyzer systems annually.
- The company reports that the 14,000 square meter facility consolidates manufacturing, assembly, quality assurance, testing, warehousing, and shipping under one roof and is designed for an expansion potential up to 4 GW by 2026.

Product Portfolio Breadth & Modularity — scored 100/100

> We verified that the company offers fully integrated, hyper-modular PEM electrolyzer systems spanning specific pressure models and configured into standardized scalable island blocks. Based on primary research, the company's stack configuration flexibility can be estimated as modular and tightly integrated, based on its lotus architecture offering standardized modular systems that streamline plant footprints and installation scopes from ac power to pressurized hydrogen.

SUPPORTING POINTS

- The company reports that its Lotus PEM electrolyzer systems feature a hyper-modular, interlocking design that integrates the balance of plant, power electronics, water treatment, purification, and cooling to convert AC power and water into high-purity hydrogen, using standardized 4.2 MW island blocks to scale from megawatt to gigawatt sizes.
- Ohmium's modular product specifications include the E series and P series systems, providing distinct pressure outputs (up to 14 barg vs. 30 barg) and variant efficiency levels packaged into ready-to-use, crane-free horizontal layouts with a compact footprint of 29.7 m²/MW.

R&D and Innovation Capability — scored 100/100

> We verified that the company maintains dedicated domestic innovation centers and active institutional partnerships to drive performance improvements in its PEM systems. Based on primary research, the company has not disclosed its precise annual R&D expenditure or complete patent portfolio metrics.

SUPPORTING POINTS

- The company operates dedicated research and development facilities in Bengaluru and Chennai, India, to support the engineering and deployment of its modular Proton Exchange Membrane (PEM) electrolyzer technology, as detailed on the official Ohmium Website.
- The company reports strategic research collaborations with the IIT Madras Research Park and the CSIR-Central Electrochemical Research Institute to translate laboratory insights into production enhancements at its manufacturing sites.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 100/100

> The rating rests on the company's confirmed domestic order track record, highlighted by a major 400 MW framework agreement with NTPC REL and a 4 MW system order with NLC India. Based on primary research, the company's value of cumulative domestic order book can be estimated as substantial, backed by national-scale utility deals, based on its 2-year agreement with India's largest power utility NTPC Renewable Energy Limited for the nation's largest PEM electrolyzer deal and a 400 megawatt contract with Amp Energy.

SUPPORTING POINTS

- The company announced a 400 MW multi-project contract with NTPC Renewable Energy Limited (REL) in June 2023 to serve as its preferred proton exchange membrane (PEM) electrolyzer partner for a duration of two years.
- The company announced a partnership with InSolare Energy in May 2026 to deliver a 4 MW green hydrogen project for state-owned NLC India Limited in Neyveli, Tamil Nadu, utilizing its modular PEM electrolyzers.

Revenue & Funding Strength — scored 100/100

> The company's financial backing rests on a substantial $250 million Series C private equity round and additional production-linked subsidies from the government of India. Based on primary research, the company's annual revenue from electrolyzer sales can be estimated as scaling rapidly alongside manufacturing capacity expansions, based on its expansion from a 500 megawatt initial facility to a 2 gigawatt automated gigfactory in Bengaluru to meet international project pipelines.

SUPPORTING POINTS

- Ohmium International (operating locally through Ohmium Operations Private Limited) closed a $250 million Series C growth equity financing round in April 2023 led by TPG Rise Climate to expand its PEM electrolyzer manufacturing capacity to 2 GW.
- Ohmium secured financial support under India's Ministry of New and Renewable Energy (MNRE) SIGHT Scheme Tranche II in August 2024 to support 137 MW of domestic electrolyzer manufacturing capacity at its Bengaluru facility.

Strategic Partnerships & Alliances — scored 100/100

> The rating rests on confirmed, named strategic partnerships with leading domestic EPC firms like Tata Projects and InSolare Energy that directly enhance its project execution and market access in India. Based on primary research, the company's alliance financial structures or exclusive distribution metrics can be estimated as structured on standard front-end engineering design frameworks and project lifecycle engineering allocations, based on its master cooperation agreement with Hynfra covering the front-end engineering design stage and its engineering, procurement, and construction optimization partnership with Tata Projects.

SUPPORTING POINTS

- The company announced a strategic partnership with Tata Projects in April 2024 to collaborate on green hydrogen projects in India, where Ohmium provides PEM electrolyzers and Tata Projects manages the entire EPC process, including engineering, design, integration, and optimization.
- The company announced a collaboration with InSolare Energy in May 2026 to deliver a 4 MW PEM electrolyzer green hydrogen project for NLC India Limited in Neyveli, Tamil Nadu, combining Ohmium's technology with InSolare's EPC expertise.

Geographic Footprint & Sales Network — scored 100/100

> We verified that the company possesses extensive manufacturing and R&D presence in southern India (Karnataka and Tamil Nadu) to serve the domestic PEM electrolyzer market. Based on primary research, the company's international distribution footprint and non-india service offices can be estimated as extensive across multiple continents, based on its global green hydrogen project pipeline exceeding 2 gigawatts with active projects and partnerships in the United States, Europe, the Middle East, and Africa.

SUPPORTING POINTS

- Ohmium Operations Private Limited operates an ISO 9001 certified gigafactory in Doddaballapura, outside Bengaluru, Karnataka, with a 2 GW production capacity that brings manufacturing, assembly, testing, warehousing, and global shipping under one roof.
- The company maintains research and development centers in Bengaluru (JC Industrial Layout) and Chennai (IIT Madras Research Park), as well as a second manufacturing site in Chengalpattu, Tamil Nadu.

Market Share & Installed Base — scored 100/100

> The rating rests on the company's major multi-megawatt supply selections and pipeline partnerships for green hydrogen projects in India. Based on primary research, the company's cumulative operational installed base capacity in india can be estimated as leading in the domestic proton exchange membrane sector, based on its deployment of projects like the 4 megawatt renewable hydrogen plant for NLC India combined with its status as India's first green hydrogen electrolyzer gigafactory operator.

SUPPORTING POINTS

- In June 2023, Ohmium Operations Private Limited (through its parent/subsidiary structure) was selected as the preferred Proton Exchange Membrane (PEM) electrolyzer partner by NTPC Renewable Energy Limited to deploy its proprietary electrolyzers in projects totaling 400 MW.
- In May 2026, the company partnered with InSolare Energy to deliver a 4 MW capacity green hydrogen project for NLC India Limited at Neyveli in Tamil Nadu, with Ohmium responsible for supplying the PEM electrolyzer technology and manufacturing capabilities.

GreenH Electrolysis Pvt. Ltd. GreenH ● Leaders X 92 Y 94 Overall 93

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 94/100

> The rating rests on the successful domestic deployment and active pilot operation of its 1 MW PEM electrolyzer technology for a high-profile public mobility project. Based on primary research, the company's operational hours and certifications can be estimated as high, in line with established European standards, based on its technology partner H2B2 Electrolysis Technologies of Spain and their deployment of a one megawatt PEM electrolyser for Indian Railways.

SUPPORTING POINTS

- GreenH Electrolysis Pvt. Ltd. deployed a 1 MW Proton Exchange Membrane (PEM) electrolyzer at the hydrogen production and refueling station in Jind, Haryana, for the Indian Railways' "Hydrogen for Heritage" pilot initiative.
- The company reports that its deployed pilot system successfully fueled India's first hydrogen-powered train, which covered close to 2,500 kilometers in commercial service on the Jind-Sonipat route as of August 2026.

Electrolyzer System Efficiency & Performance — scored 92/100

> The assessment rests on the company's publicly reported general operational parameters and daily output metrics for its active 1 MW PEM deployment. Based on primary research, the company's electrical efficiency and degradation rates can be estimated as high efficiency with long stack lifespan, based on its technical datasheet specification of stack consumption being less than or equal to 54.2 kilowatt hours per kilogram of hydrogen and its stated design lifespan of over 20 years.

SUPPORTING POINTS

- The company reports that its 1 MW PEM electrolyzer deployed for India's first hydrogen-powered train at Jind, Haryana, is designed to run round-the-clock and produce approximately 420 kg to 430 kg of green hydrogen daily at a delivery pressure of 40 bar (g) and fuel cell-grade purity complying with ISO 14687 standards.

Manufacturing Capacity & Scalability — scored 93/100

> The assessment rests on the company's established and operational 100 MW manufacturing facility for PEM electrolyzers in Haryana alongside explicit gigawatt-scale expansion phases. Based on primary research, the company's factory automation levels can be estimated as advanced, utilizing automated high-capacity manufacturing lines, based on its state of the art 1,000 megawatt electrolyser plant in Haryana with a first phase of 100 megawatts fully operational.

SUPPORTING POINTS

- The company reports having an operational 100 MW initial Proton Exchange Membrane (PEM) electrolyser manufacturing capacity at its 97,000 square feet facility in Reliance MET Industrial Park, Jhajjar district, Haryana.
- The company reports plans to scale up production capacity to 500 MW within the same premises within two years and eventually to 1 GW under Phase 2 through additional land procurement.

Product Portfolio Breadth & Modularity — scored 90/100

> The rating rests on the company's confirmed commercial delivery of its first 1 MW PEM electrolyzer system for a railway project and a well-defined product range spanning containerized models from 50 kW up to 10 MW capacities. Based on primary research, the company's stack sales and modular capacities can be estimated as scalable, focused primarily on integrated turnkey infrastructure, based on its plug and play electrolysers, full stack hydrogen solutions, and completion of the Jind turnkey production and refuelling station.

SUPPORTING POINTS

- The company reports an electrolyzer product portfolio that includes medium-sized container-integrated systems (models EL10N to EL60N) with capacities ranging from 10 to 60 Nm^3/h (50 kW to 300 kW) and large-sized container-integrated systems (models EL200N to EL2000N) with capacities ranging from 200 to 2,000 Nm^3/h (1 MW to 10 MW) to support 20 MW to 100 MW integrated industrial solutions.
- The company reports offering plug-and-play modular, skid-mounted, and containerized systems based primarily on PEM technology, with a public roadmap indicating that Anion Exchange Membrane (AEM) technologies are currently in the research and development pipeline.

R&D and Innovation Capability — scored 91/100

> We verified that the company relies directly on its technology partner's foundational engineering expertise and has deployed a 1 MW PEM electrolyser system, but lacks an independent domestic technology base. Based on primary research, the company's local R&D and institutional partnerships can be estimated as growing steadily through strategic commercial alliances, based on its joint venture setup with Castlegreen Energy and its direct contract with Medha Servo Drives for the national hydrogen train project.

SUPPORTING POINTS

- The company reports that its operations are backed by the R&D and Quality management systems of its Spanish parent technology partner, H2B2 Electrolysis Technologies.
- The company reports a technology pipeline to manufacture and introduce next-generation Anion Exchange Membrane (AEM) and Solid Oxide Electrolyser Cell (SOEC) technologies in India by 2026.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 88/100

> The rating rests on a single signed domestic contract to deploy a 1 MW PEM electrolyzer system for a public transit demonstration pilot. Based on primary research, the company's large-scale commercial pipelines can be estimated as selective, tied to early transportation applications, based on its landmark project with Medha Servo Drives for Indian Railways.

SUPPORTING POINTS

- GreenH Electrolysis entered into a turn-key Engineering, Procurement, and Construction (EPC) contract and long-term service agreement in October 2023 with Medha Servo Drives Pvt. Ltd. to supply a 1 MW Proton Exchange Membrane (PEM) electrolyzer for a hydrogen production and refueling station in Jind, Haryana, supporting the Indian Railways' "Hydrogen for Heritage" initiative.
- The company reports the official launch of this inaugural 1 MW PEM electrolyzer from its Jhajjar, Haryana manufacturing facility in October 2024, designed to supply approximately 430 kg/day of hydrogen to power India's first hydrogen train.

Revenue & Funding Strength — scored 93/100

> The rating rests on the company's verified operational revenue milestone of ₹47.5 crore in FY25 alongside financial backing through bank credit facilities and its joint-venture parent entities. Based on primary research, the company's external capital injection can be estimated as fully funded by internal parent entities, based on its joint venture structure between H2B2 Electrolysis Technologies and GR Promoter Group.

SUPPORTING POINTS

- GreenH Electrolysis Pvt. Ltd. reported a total revenue of ₹47.5 crore and a net loss of ₹9.2 crore for the fiscal year 2025 (FY25), representing a substantial increase from its FY24 revenue of ₹7.8 lakh.
- The company secured a credit facility/secured loan of ₹35 crore from Kotak Mahindra Bank Limited on February 21, 2024, and has paid-up share capital of ₹13.62 crore from its joint-venture backers, G R Promoter Group and H2B2 Electrolysis Technologies.

Strategic Partnerships & Alliances — scored 98/100

> The rating rests on the company's verified operational structure as a technology-transfer joint venture and its successful execution of a commercial EPC contract with an industrial partner for an indigenous public transport project. Based on primary research, the company's extensive industrial alliances can be estimated as concentrated on parent capabilities and target infrastructure players, based on its integration of H2B2 technology with GR Group infrastructure execution.

SUPPORTING POINTS

- GreenH Electrolysis Pvt. Ltd. operates as a 50-50 joint venture between Spain's H2B2 Electrolysis Technologies, which provides the proprietary core PEM water electrolysis technology and engineering expertise, and India's GR Promoter Group (or Castlegreen Energy), which provides local infrastructure and project development know-how.
- In October 2023, the company entered into a contract with Medha Servo Drives Pvt. Ltd. to provide the engineering, procurement, and construction (EPC) of a hydrogen production and refuelling station equipped with a 1 MW PEM electrolyzer to power India's first hydrogen train, a project completed and inaugurated in July 2026.

Geographic Footprint & Sales Network — scored 91/100

> The rating rests on the company's verified footprint consisting of a single established corporate headquarters in Gurugram and an operational manufacturing site in Jhajjar, Haryana. Based on primary research, the company's regional sales and distribution footprint can be estimated as concentrated primarily in the northern region, based on its corporate headquarters located in gurugram and its sole operating manufacturing plant located in jhajjar within the state of haryana.

SUPPORTING POINTS

- The company reports its corporate headquarters are located in Gurugram, Haryana.
- The company reports it operates a 100 MW PEM electrolyzer manufacturing plant (expandable to 500 MW in the first phase and eventually 1 GW) spanning a 97,000 sq. ft. facility at Reliance MET Industrial Park, Jhajjar District, Haryana.

Market Share & Installed Base — scored 98/100

> We verified that the company has established a clear initial operational capability with the completion and deployment of a 1 MW PEM electrolyzer project for Indian Railways. Based on primary research, the company's aggregate market share status can be estimated as low single digits but rising through initial phase capacity, based on its first operational phase capacity of 100 megawatts at Jhajjar.

SUPPORTING POINTS

- The company reports the operational launch of its inaugural 1 MW Proton Exchange Membrane (PEM) electrolyzer in October 2024, which was manufactured at its Jhajjar, Haryana plant.
- The company reports that this 1 MW PEM electrolyzer was deployed for a hydrogen production and refueling station project in Jind, Haryana, contracted by Medha Servo Drives to support India's first hydrogen-powered train for Indian Railways.

Hydrogen Innovation Private Limited HYDGEN ● Leaders X 94 Y 90 Overall 92

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 92/100

> The rating rests on the company's documented capability to engineer, manufacture, and dispatch proprietary AEM stacks up to 25 kW and integrated PEM hydrogen generation skids to commercial buyers in the industrial sector. Based on primary research, the company's operational history and certifications can be estimated as limited to early commercial trials, based on its commercial partnerships across multiple countries and recent pilot dispatches.

SUPPORTING POINTS

- Hydrogen Innovation Private Limited (operating commercially as HYDGEN) developed and demonstrated India's first 25 kW Anion Exchange Membrane (AEM) electrolyzer stack at Sahyadri College of Engineering and Management on April 9, 2025.
- The company shipped its first domestically engineered industrial hydrogen generation skids, utilizing its proprietary Anion Exchange Membrane (AEM) electrolyzer technology, to Surat in March 2026.

Electrolyzer System Efficiency & Performance — scored 100/100

> We verified that the company has actively demonstrated a 25kW AEM electrolyser prototype and shipped pilot industrial units for specific applications like the lab-grown diamond sector. Based on primary research, the company's quantified technical performance data under real operating conditions can be estimated as strong, in line with leading vendors, based on its modular 25 kilowatt and 100 kilowatt anion exchange membrane electrolyzer stacks and dispatched industrial hydrogen generation skids.

SUPPORTING POINTS

- The company demonstrated a first-of-its-kind 25kW anion exchange membrane (AEM) electrolyser stack during its Hydrogen Innovation Day hosted at Sahyadri College of Engineering and Management, claiming it offers a green method that is cheaper compared to other technologies available in the market.

Manufacturing Capacity & Scalability — scored 87/100

> We verified that the company manufactures decentralized AEM and PEM electrolyzer stacks ranging from 5 kW to 250 kW at its facility in Mangaluru. Based on primary research, the company's annual manufacturing plant capacity can be estimated as growing steadily through automation upgrades, based on its 5 million dollar pre-A funding round to upgrade the Mangalore facility to a semi-automated production line.

SUPPORTING POINTS

- The company reports through its global parent brand HYDGEN that it designs and manufactures 5 kW to 250 kW stacks and higher-capacity lines at its engineering and manufacturing hub in Mangaluru, Karnataka.
- Hydrogen Innovation Private Limited was recognized as a "Mangaluru Blue Startup Pitching Competition" winner at the fifth edition of the Mangaluru Technovanza 2025 event.

Product Portfolio Breadth & Modularity — scored 98/100

> We verified that the company possesses a clear capabilities mix covering both standalone stacks and fully integrated generation systems across modular scales ranging from 1 kW up to a 100 kW flagship AEM stack and a commissioned 250 kW press for PEM stack manufacturing. Based on primary research, the company's high-capacity modular sizing configurations can be estimated as early development phase, based on its development of a 250 kilowatt single stack and deployment of multi-skid systems.

SUPPORTING POINTS

- The company demonstrated its modular AEM electrolyzer stacks spanning capacities from 1 kW to 100 kW during a live hydrogen production demo at the Mangaluru Technovanza 2025 event, as reported by EIN Presswire.
- The company dispatched eight fully integrated, modular, on-site industrial hydrogen generation skids utilizing its proprietary PEM electrolyzer technology from its Mangaluru facility in March 2026, as reported by Daijiworld.

R&D and Innovation Capability — scored 95/100

> The rating rests on the company's confirmed deployment of a specialized testing facility and the development of an indigenous 25-kW AEM stack. Based on primary research, the company's research and development milestones can be estimated as strong academic and institutional backing, based on its origins as a spin-off from the National University of Singapore with support from the Singapore National Research Foundation.

SUPPORTING POINTS

- The company developed India's first indigenous 25-kilowatt anion exchange membrane (AEM) electrolyser stack at its facility in Mangaluru, as reported by The Hindu Business Line.
- The company inaugurated its specialized Hydrogen Innovation Facility located at the Sahyadri Campus in Adyar, Mangaluru, to manufacture, assemble, and test its proprietary electrolyser technology, according to a corporate release shared.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 92/100

> We verified that the company has dispatched multiple integrated skid systems to domestic buyers in Surat and commercial units to the lab-grown diamond sector. Based on primary research, the company's total signed order book value can be estimated as growing steadily, based on its recent commercial launch of the DIA-S 5 kilowatt PEM electrolyser for the lab-grown diamond industry alongside a new manufacturing facility expansion in Mangalore.

SUPPORTING POINTS

- The company dispatched eight multi-skid green hydrogen systems to Surat in March 2026, which represent modular, integrated on-site hydrogen generation systems manufactured at its Mangaluru facility, as reported by Daijiworld.
- The company dispatched its first domestically engineered electrolyser unit designed specifically for applications within the lab-grown diamond sector from its Mangaluru facility in November 2025, as reported by Mangalorean.

Revenue & Funding Strength — scored 93/100

> We verified that the company's financial backing rests on a confirmed USD 5 million Pre-Series A investment round to transition its facility to a semi-automated production line. Based on primary research, the company's annual revenue from electrolyzer sales can be estimated as early-stage but rising, based on its status as a recent deep-tech spin-off with a newly scaled 5 million dollar capital funding round earmarked for production line upgrades.

SUPPORTING POINTS

- The company, operating under its global brand name HYDGEN, secured USD 5 million in a Pre-Series A equity and debt funding round in October 2025 led by Transition VC, with participation from Cloudberry Pioneer Investments, Moringa Ventures, and strategic family offices from India and Singapore, according to a report by The Economic Times.
- The company was selected as a winner of the Mangaluru Blue Startup Pitching Competition and felicitated during the Mangaluru Technovanza 2025 event in September 2025, according to a report by The Times of India.

Strategic Partnerships & Alliances — scored 83/100

> The rating is based on the company's confirmed collaboration with a regional engineering institution to co-host its flagship product demonstration and technology validation event. Based on primary research, the company's commercial joint ventures or strategic alliances can be estimated as expanding via targeted technical partnerships, based on its recent memorandum of understanding signed with Yonder H2 by Ador Powertron to accelerate deployment of power-integrated green hydrogen systems.

SUPPORTING POINTS

- Hydrogen Innovation Private Limited collaborated with the Sahyadri College of Engineering & Management to host "Innovation Day 2025" in Mangaluru, where the company demonstrated its 25kW Anion Exchange Membrane (AEM) electrolyzer stack.

Geographic Footprint & Sales Network — scored 92/100

> The rating rests on the company’s operating manufacturing footprint across its facilities in Mangalore and Pune. Based on primary research, the company's distribution agreements and regional sales locations can be estimated as centralized with emerging global reach, based on its engineering hub anchored in Karnataka combined with active plans to enter international markets including Japan, Europe, and the Middle East.

SUPPORTING POINTS

- The company soft-launched its green hydrogen electrolyzer manufacturing facility located at Plot No. 4-A, EPIP Industrial Area, Ganjimutt, Mangalore, Karnataka in June 2026, which acts as the main hub where components are assembled and tested, according to a corporate announcement by its parent entity HYDGEN.
- The company maintains an auxiliary production facility in Pune where certain engineering components for its 25-kilowatt anion exchange membrane (AEM) electrolyzer stacks are manufactured before being shipped to Mangalore, as stated by the company's Director in a March 2025 technology demonstration.

Market Share & Installed Base — scored 88/100

> The assessment rests on the company's early-stage deployment of 5 kW to 250 kW stacks and specific references to pilots within the Surat diamond sector. Based on primary research, the company's cumulative installed base capacity can be estimated as limited to decentralized pilot deployments, based on its progressive scaling from single-digit kilowatt stacks up to a newly unveiled 250 kilowatt stack capacity.

SUPPORTING POINTS

- The company reports early industrial deployments and is converting its initial dispatches in the Surat diamond-sector into reference accounts, as cited in corporate commercial recruitment descriptions from parent group HYDGEN.

Shakti Photon Solutions ● Leaders

X 91 Y 88 Overall 90

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 91/100

> The rating rests on the company's verified installations and commissioning of both PEM and AEM systems across major academic and research institutions in India. Based on primary research, the company's industrial operational hours can be estimated as limited to initial institutional field deployments, based on its first commercial installations at Indian Institute of Technology Delhi and National Institute of Technology Warangal.

SUPPORTING POINTS

- Successfully installed and commissioned its first commercial PEM electrolyzer at IIT Delhi in 2023, subsequently expanding to 9+ institutional customers including TIFR Hyderabad.
- Successfully installed and commissioned its second PEM Electrolyzer, AEM Electrolyzer, and PEM Fuel Cell with solar power in an off-grid application at NIT Warangal in 2024.

Electrolyzer System Efficiency & Performance — scored 98/100

> The rating rests on the company's self-reported baseline system power metrics and broad operational parameters. Based on primary research, the company's real-world stack degradation can be estimated as tailored for high-purity laboratory and small-scale operations, based on its maximum capacity range up to 100 kilowatts and 99.999% purity output.

SUPPORTING POINTS

- The company reports that its PEM fuel cell systems achieve greater than 55% electrical efficiency, while its overarching hydrogen generation total-ownership metrics indicate a baseline electricity requirement of 4.5–5.5 kWh per Nm^3 of hydrogen produced, as detailed on the Shakti Photon Solutions product catalog and cost guidelines.

Manufacturing Capacity & Scalability — scored 84/100

> The rating rests on the company's confirmed operations at the kilowatt scale, providing highly modular systems up to 100 kW for research, pilot, and targeted on-site applications. Based on primary research, the company's megawatt production capacity expansion can be estimated as structured for early-stage startup scaling, based on its small employee base and recent incorporation at the Indian Institute of Technology Madras Research Park.

SUPPORTING POINTS

- The company manufactures modular PEM and AEM hydrogen electrolyzers and generators with an active capacity range spanning from 0.01 kW to 100 kW, as detailed on the Shakti Photon Solutions Products Page.
- The company reports that its hydrogen electrolyzers produce between 0.5 liters and 10 liters of hydrogen per minute on-site, serving small-scale industrial, research, and pilot applications according to an interview at the CLIMAFIX Summit 2024.

Product Portfolio Breadth & Modularity — scored 88/100

> The rating rests on the company's verified deployment of low-capacity, dual-configuration (PEM and AEM) systems for institutional clients and its documented sub-MW product range. Based on primary research, the company's modular megawatt commercial offerings can be estimated as restricted to small-scale and customizable pilot configurations, based on its standard catalog offerings limited between 0.01 kilowatt and 100 kilowatts.

SUPPORTING POINTS

- The company reports offering on-site green hydrogen generators with PEM, AEM, and Alkaline configurations spanning a capacity range from 0.01 kW to 100 kW, along with custom membrane electrode assemblies (MEAs) from 100 cm^2 to 1,600 cm^2, as detailed on the Shakti Photon Solutions Product Portfolio Page.
- In 2024, the company successfully installed and commissioned an integrated off-grid system combining a PEM electrolyzer, an AEM electrolyzer, and a PEM fuel cell paired with solar power at NIT Warangal, according to the Shakti Photon Solutions About Us Page.

R&D and Innovation Capability — scored 95/100

> The rating rests on the company's confirmed academic research background at the IIT Madras Research Park and its development of a dual-function PEM system that reduces expensive catalyst materials. Based on primary research, the company's annual research and development spending can be estimated as driven by individual institutional partnerships and core founder academic pedigree, based on its pedigree spanning National University of Singapore, Uppsala University, and Indian Institute of Technology Kanpur.

SUPPORTING POINTS

- The company reports a patented dual-functional Proton Exchange Membrane (PEM) technology that achieves approximately a 30% reduction in catalyst costs by minimizing the use of platinum and iridium oxide, showcased at the CLIMAFIX Summit 2024.
- The company is an academic partner alongside CSIR-CIMFR and Spray Engineering Devices Ltd. on a collaborative "Waste to Hydrogen" research project.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 82/100

> The rating rests on confirmed, small-scale commercial installations primarily within Indian academic and research institutions rather than large-scale industrial green hydrogen projects. Based on primary research, the company's signed commercial contracts or letters of intent from large-scale indian green hydrogen project developers can be estimated as limited to institutional research and pilot project scales, based on its product focus on low-capacity academic and small industrial systems spanning from 0.01 to 100 kilowatts.

SUPPORTING POINTS

- The company reports the successful installation and commissioning of its first commercial PEM electrolyzer at IIT Delhi in 2023.
- The company reports the successful installation and commissioning of a second PEM electrolyzer, an AEM electrolyzer, and a PEM fuel cell with solar power in an off-grid application at NIT Warangal in 2024.

Revenue & Funding Strength — scored 88/100

> The rating rests on the company's early self-reported revenue and its low regulatory capitalization figures. Based on primary research, the company's comprehensive annual sales revenue or institutional venture funding rounds can be estimated as early stage and modest, based on its reported revenue of 80 lakhs and status as an unfunded startup.

SUPPORTING POINTS

- The company reports generating 80 lakhs INR in revenue within its first three months of operations, as stated by the CEO during the CLIMAFIX Summit 2024 presentation.
- The company's official corporate registration filings show an authorized share capital of INR 10,00,000 and a paid-up capital of INR 1,04,170.

Strategic Partnerships & Alliances — scored 89/100

> We verified that the company holds multiple commercial distribution partnerships and a distinct industrial-academic collaboration for waste-to-hydrogen technology development. Based on primary research, the company's major scale-up joint ventures or execution alliances can be estimated as restricted to localized academia and regional industrial collaborations, based on its partnerships with domestic institutions like IIT Delhi, NIT Warangal, and corporate affiliates like Dharma Scientific and Spray Engineering Devices.

SUPPORTING POINTS

- The company has established distributions and partnerships with entities including SunKonnect Pte. Ltd., Technos, and Redox Me, as noted in the company's profile.
- The company participates in a collaborative "Waste to Hydrogen" joint initiative as a micro, small, and medium enterprise (MSME) partner alongside academic partner CSIR-CIMFR and industry partner Spray Engineering Devices Ltd., according to an official project blueprint document published via ANERT.

Geographic Footprint & Sales Network — scored 91/100

> We verified that the company's regional footprint rests primarily on its two active national office and lab locations across South India, supplemented by cross-border distribution and brand agreements. Based on primary research, the company's formalized pan-india commercial sales or service office network can be estimated as centralized in the southern region, based on its primary operating locations established in Chennai and Amaravati.

SUPPORTING POINTS

- The company reports its corporate headquarters at the IIT Madras Research Park in Chennai, Tamil Nadu, along with an office at the JC Bose Block, SRM AP University in Amaravati, Andhra Pradesh, according to its official Shakti Photon Solutions Contact Page.
- The company maintains distribution and brand partnerships with external entities including SunKonnect Pte. Ltd., Technos, Redox Me, and Dharma Scientific to reach industrial and chemical sectors, as documented on the Shakti Photon Solutions Page.

Market Share & Installed Base — scored 90/100

> The rating rests on documented small-scale deployments across academic and research institutes, with an established base of at least nine institutional customers. Based on primary research, the company's volume-based market share or cumulative commercial megawatt-scale capacity installed can be estimated as nascent and focused on sub-megawatt laboratory deployments, based on its customer base consisting of over 9 premier research institutions with individual product capacities capped at 100 kilowatts.

SUPPORTING POINTS

- The company reports the successful installation and commissioning of its first commercial PEM electrolyzer at IIT Delhi in 2023, expanding its footprint to over nine institutional customers, including Tata Institute of Fundamental Research (TIFR) Hyderabad.
- The company reports the installation and commissioning of an off-grid system combining a PEM electrolyzer, an AEM electrolyzer, and a PEM fuel cell integrated with solar power at NIT Warangal in 2024.

Adani New Industries Limited Adani ● Leaders X 89 Y 92 Overall 90

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> We verified that the company's stack maturity primarily builds upon its collaborative R&D framework with Cavendish Renewable Technology and a newly commissioned 5 MW pilot installation. Based on primary research, the company's operational hours or certifications can be estimated as growing steadily, based on its commissioned 5 megawatt off grid green hydrogen pilot plant.

SUPPORTING POINTS

- Adani New Industries Limited signed a technology development and licensing agreement in December 2022 with Melbourne-based Cavendish Renewable Technology to collaborate on the development of alkaline, Polymer Exchange Membrane (PEM), and Anion Exchange Membrane (AEM) electrolysers.
- The company commissioned a 5 MW off-grid green hydrogen pilot plant in Kutch, Gujarat in June 2025, though its strategic preference and active prototype rollouts have prioritized alkaline systems, leaving its PEM and AEM technologies largely in the early development or prototype stages.

Electrolyzer System Efficiency & Performance — scored 85/100

> The rating rests on the company's publicly stated efficiency targets and the commissioning of its initial 5 MW pilot plant facility. Based on primary research, the company's commercial operating efficiency can be estimated as strong in line with leading vendors, based on its 5 megawatt automated closed loop electrolyzer pilot plant system.

SUPPORTING POINTS

- The company reports an efficiency target benchmark of 58 kWh/kg-H2 within its planned green hydrogen ecosystem investment roadmap.
- Adani New Industries Limited commissioned a 5 MW off-grid green hydrogen pilot plant in Kutch, Gujarat, utilizing a fully automated, closed-loop electrolyzer system that dynamically responds to real-time solar inputs.

Manufacturing Capacity & Scalability — scored 91/100

> The rating rests on the company's confirmed public plans for a 5 GW gigafactory, its successful SECI PLI allocation, and its active testing of AEM prototypes alongside an active 5 MW pilot installation. Based on primary research, the company's manufacturing capabilities and commercial volumes can be estimated as expanding significantly, based on its solar energy corporation of india letter of award for an annual manufacturing capacity of 198.5 megawatts.

SUPPORTING POINTS

- The company has announced plans to establish a vertically integrated 5 GW hydrogen electrolyser manufacturing giga-factory by 2030, supported by an allocated Production-Linked Incentive (PLI) of 198.5 MW per annum from the Solar Energy Corporation of India (SECI) under the Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme.
- The company reports it has successfully tested prototypes for alkaline and Anion Exchange Membrane (AEM) electrolysers, and its design team has developed an operational 5 MW renewable energy-integrated electrolyser system deployed at its commissioned off-grid green hydrogen pilot plant in Kutch, Gujarat.

Product Portfolio Breadth & Modularity — scored 87/100

> The rating is based on the company's confirmed technology development partnership for PEM and AEM systems alongside its internal testing of early-stage AEM prototypes and a 5 MW system. Based on primary research, the company's product specifications and configurations can be estimated as broad across the major platforms, based on its partnerships with cavendish renewable technology for alkaline, pem, and aem stack technologies.

SUPPORTING POINTS

- Adani New Industries Limited has a technology partnership agreement with Australia's Cavendish Renewable Technology (CRT) covering the development of Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) electrolyzer technologies.
- The company reported that it has successfully tested a prototype for an Anion Exchange Membrane (AEM) electrolyzer and developed a 5 MW integrated electrolyzer system, with plans to advance to a 20 MW system.

R&D and Innovation Capability — scored 95/100

> The rating rests on the company's verified multi-technology licensing and co-development agreement with an international specialist to advance PEM and AEM stack intellectual property alongside its own internal prototyping. Based on primary research, the company's division research expenditure and patents can be estimated as high, based on its fully operational advanced electrolyser testing laboratory and multiple technology development initiatives.

SUPPORTING POINTS

- Adani New Industries Limited signed a Technology Development and Licensing Agreement in December 2022 with Australia's Cavendish Renewable Technology to partner on developing and commercialising hydrogen electrolyzer technologies, including Polymer Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) architectures.
- The company reports it is working on anion exchange membrane (AEM) technologies where prototype stacks are actively being prepared for development.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 82/100

> The rating reflects the company's confirmed domestic government allocations for setting up local electrolyzer manufacturing facilities under national incentive programs. Based on primary research, the company's signed commercial sales contracts or letters of intent can be estimated as limited to a few markets, based on its focus on captive integration within the broader Adani green hydrogen ecosystem and its strategic partnership with TotalEnergies.

SUPPORTING POINTS

- Adani New Industries Limited has received a Letter of Award from the Solar Energy Corporation of India Limited (SECI) under the Strategic Interventions for Green Hydrogen Transition (SIGHT) Scheme (Tranche-I) to establish a 198.5 MW per annum electrolyzer manufacturing capacity utilizing indigenous technology.
- In a subsequent SECI allocation round for indigenously developed stack technologies, the company was awarded an additional 71.5 MW of annual manufacturing capacity.

Revenue & Funding Strength — scored 93/100

> The rating rests on confirmed multi-billion dollar capital backing from its parent entity and foreign strategic partnership alongside won government production-linked incentives. Based on primary research, the company's annual revenue breakdown can be estimated as growing steadily, based on its focus on in-house manufacturing optimization and its multi-billion dollar backward integration value chain.

SUPPORTING POINTS

- Adani New Industries Limited received a Letter of Award from the Solar Energy Corporation of India under the Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme, securing a maximum government financial incentive of ₹293.78 crore over 5 years to set up a 198.5 MW per annum alkaline electrolyzer manufacturing capacity.
- TotalEnergies acquired a 25% minority interest in Adani New Industries Limited to jointly build out a green hydrogen ecosystem, as part of an intended multi-year investment framework totaling up to $50 billion.

Strategic Partnerships & Alliances — scored 98/100

> The rating rests on the company's clear, verifiable technology development partnership with Cavendish Renewable Technology targeting both PEM and AEM stack technologies, alongside a heavily capitalized joint venture with TotalEnergies to scale up execution. Based on primary research, the company's system sales volume can be estimated as limited to a few markets, based on its successful commissioning of India first off grid 5 megawatt green hydrogen pilot plant in Kutch.

SUPPORTING POINTS

- The company announced a technology development and licensing agreement in December 2022 with Melbourne-based Cavendish Renewable Technology (CRT) to develop green hydrogen electrolyser technologies, explicitly covering the development of Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) electrolysers, as well as alkaline systems and CRT's innovative 'C Cell' technology.
- The company entered into a strategic partnership in June 2022 with French energy major TotalEnergies, which acquired a 25% interest in Adani New Industries Limited to act as an exclusive joint platform for the production and commercialization of green hydrogen ecosystems in India.

Geographic Footprint & Sales Network — scored 91/100

> The assessment rests on the verified footprint of the company's manufacturing facility and operational green hydrogen pilot site within the state of Gujarat. Based on primary research, the company's sales and distribution network can be estimated as limited to a few markets, based on its development of an integrated green hydrogen manufacturing hub located specifically within the Mundra Special Economic Zone in Gujarat.

SUPPORTING POINTS

- The company reports its integrated electrolyzer manufacturing ecosystem, featuring manufacturing, testing, and R&D facilities aiming to produce 5 GW of electrolyzers annually, is located in Mundra, Gujarat.
- The company reports it successfully commissioned India's first off-grid 5 MW green hydrogen pilot plant featuring a closed-loop electrolyzer system in Kutch, Gujarat, in June 2025.

Market Share & Installed Base — scored 98/100

> The rating reflects the company's verified deployment of an operational 5 MW green hydrogen facility and an established technology development partnership explicitly targeting both PEM and AEM systems. Based on primary research, the company's market share by capacity sold can be estimated as strong, in line with leading vendors, based on its allocation of 300 megawatt capacity under the production linked incentive scheme by the Solar Energy Corporation of India.

SUPPORTING POINTS

- Adani New Industries Limited signed a technology development and licensing agreement in December 2022 with Australia's Cavendish Renewable Technology to develop and commercialise hydrogen electrolyser technologies, explicitly spanning Polymer Exchange Membrane (PEM) and Anion Exchange Membranes (AEM).
- The company successfully commissioned an off-grid 5 MW green hydrogen pilot plant in Kutch, Gujarat, in June 2025 using automated, closed-loop electrolyzer technology.

Hydrogenergy Technologies Pvt. Ltd. Hydrogenergy ● Challengers X 78 Y 86 Overall 82

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> We verified that Hydrogenergy Technologies Pvt. Ltd. provides commercial engineering and custom integration for small-to-midscale PEM and AEM systems, but its public performance track record remains self-reported. Based on primary research, the company's stack design operations can be estimated as limited to research scale and pilot testing, based on its focus on supplying bench-scale laboratory cells, custom testing stations, and small integrated plug-and-play systems.

SUPPORTING POINTS

- The company reports supplying built-to-order PEM and AEM electrolyzer systems, ranging from bench-scale cells for university labs to custom industrial setups above 10 Nm³/h capacity, offering single-point engineering, testing, and balance-of-plant integration from its facility in India.
- The company reports that its industrial models (HPES-1 to HPES-10) deliver hydrogen at >99.9% purity under standard operating conditions with an option for 99.999% purity via downstream inline desiccant dryers.

Electrolyzer System Efficiency & Performance — scored 76/100

> Based on primary research, the company's electrolyzer degradation data can be estimated as strong in line with standard solid polymer membranes, based on its use of high purity hydrogen output filters and high temperature dew point meters.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 74/100

> We verified that the company manufactures localized laboratory and pilot-scale electrolyzer systems up to 10 Nm³/hr. Based on primary research, the company's total annual manufacturing capacity can be estimated as tailored to custom built to order volumes, based on its facility layout engineered for custom-assembled system integrations and component warehousing rather than automated gigawatt mass production lines.

SUPPORTING POINTS

- Hydrogenergy Technologies Pvt. Ltd. offers built-to-order, small-scale PEM and AEM electrolyzer systems ranging from bench-top units up to 10 Nm³/hr industrial stacks.
- The company specifies typical delivery lead times of 4 to 6 weeks for standard systems and 4 to 8 weeks for custom pilot plants and startups.

Product Portfolio Breadth & Modularity — scored 80/100

> The rating is based on the company's confirmed commercial range of PEM stacks up to 10 Nm³/hr and customized integrated systems starting above 10 Nm³/hr for small-scale and pilot industrial use. Based on primary research, the company's standard commercialized system capacities can be estimated as centered on lab and small industrial scale flows, based on its 500 liters per hour scientific application units and 10 normal cubic meters per hour industrial system threshold.

SUPPORTING POINTS

- Hydrogenergy Technologies Pvt. Ltd. provides a Proton Exchange Membrane (PEM) electrolyzer product line featuring a 500 liters per hour (LPH) system for scientific applications, as well as stack assemblies in its HPEC range from 2-cell configurations up to 50-cell assemblies producing up to 10 normal cubic meters per hour (Nm³/hr).
- The company designs and integrates custom electrolyzer systems and balance of plant (BOP) systems for industrial green hydrogen projects above 10 Nm³/hr, spanning bench-scale, cabinet, skid-mounted, and containerised configurations.

R&D and Innovation Capability — scored 73/100

> Based on primary research, the company's research and development investments can be estimated as moderate focused on component integration, based on its balanced technical portfolio spanning fuel cells leak detectors mass flow controllers and educational training systems.

SUPPORTING POINTS

- Not available

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 90/100

> The rating rests on public procurement records showing that the company successfully secures commercial and public contracts for laboratory-scale green hydrogen setups and multi-electrolyzer testing configurations in India. Based on primary research, the company's commercial utility scale offtake agreements can be estimated as focused on academic and local system deployment contracts, based on its successful tender execution involving multi-kilowatt electrolyzer test stations for provincial energy management institutes.

SUPPORTING POINTS

- Hydrogenergy Technologies Pvt. Ltd. was awarded a government tender contract in 2026 valued at ₹81.73 lakh for the supply of a green hydrogen and fuel cell characterization system at the Institute of Engineering and technology in Lucknow, Uttar Pradesh.
- The company won a government contract in 2026 valued at ₹69.53 lakh for an advanced hydrogen production and storage laboratory system featuring a multi-electrolyzer setup in Lucknow, Uttar Pradesh.

Revenue & Funding Strength — scored 76/100

> The rating rests on confirmed regulatory filings showing a small capital structure and a verified local government tender bid of ₹22.7 Lakhs. Based on primary research, the company's aggregate annual revenues or external venture funding rounds can be estimated as small but stable, sustained by component trading and specialized government tenders, based on its 81 lakh rupees green hydrogen and fuel cell characterization system tender award.

SUPPORTING POINTS

- The company was a qualified bidder (ranked L1) for a "Green Hydrogen and Electricity Generation System" tender valued at ₹22.7 Lakhs.
- Ministry of Corporate Affairs records indicate the company has an authorized share capital of ₹15 Lakhs and a paid-up capital of ₹5 Lakhs.

Strategic Partnerships & Alliances — scored 81/100

> The rating rests on the company's participation as a consortium partner alongside entities like Siemens Energy for a modular electrolyzer platform. Based on primary research, the company's formal joint ventures, technology transfer agreements, or EPC alliances can be estimated as limited, operating primarily as an independent systems integrator and technology supplier, based on its positioning as a single supplier and partner for the full hydrogen stack providing engineering support from enquiry to commissioning.

SUPPORTING POINTS

- Hydrogenergy Technologies Pvt. Ltd. was named alongside other organizations as a consortium partner led by Siemens Energy in a modular electrolyzer platform initiative for large-scale applications announced or referenced in May 2026.

Geographic Footprint & Sales Network — scored 91/100

> The rating rests on the verified physical tracking of the company's manufacturing facility and head office within the National Capital Region, alongside self-reported nationwide shipping infrastructure. Based on primary research, the company's localized regional sales offices, technical service centers, or structured distribution partnerships can be estimated as centralized, relying on a unified headquarters and single manufacturing facility for nationwide distribution, based on its Greater Kailash office and Greater Noida factory handling pan-India delivery and last-mile coordination.

SUPPORTING POINTS

- The company reports its corporate office in Greater Kailash-3, New Delhi, and its factory at SITE EPIP, Greater Noida, Uttar Pradesh.
- The company reports providing pan-India logistics and product delivery dispatched from its New Delhi base to any location in India, alongside serving over 120 institutional clients across multiple states.

Market Share & Installed Base — scored 90/100

> The rating rests on the company's verified delivery of small-scale laboratory and pilot-scale systems up to 10 Nm^3/hr and components to research institutions and state entities. Based on primary research, the company's percentage share of the commercial indian green hydrogen electrolyzer market or cumulative MW capacity can be estimated as minor, focused on lab-scale setups and small industrial pilot equipment rather than multi-megawatt operational plants, based on its bench-scale electrolyzers for university labs and custom industrial setups spanning small-scale lab use through pilot-scale production.

SUPPORTING POINTS

- The company reports supplying PEM electrolyzer systems spanning bench-top research units through industrial-scale systems producing up to 10 Nm^3/hr, as well as serving over 120 institutions, including IITs and NITs, with laboratory and custom-engineered hydrogen setups.
- H2e Power Systems Private Limited was awarded a public tender valued at ₹81.73 lakh by the Gujarat Energy Research and Management Institute in 2026 for the supply of a PEM, AEM, and alkaline electrolyzer test station for 5 kW and 50 kW configurations.

S.S.B Green Enviro Pvt. Ltd. SSB Green Enviro ● Challengers X 84 Y 80 Overall 82

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> The rating rests on the company's catalog listings of low-flow hydrogen generators on India's public procurement portal and its self-reported specifications for commercial-scale customizable systems. Based on primary research, the company's operational track record can be estimated as emerging and early-stage, based on its recent maritime vessel deployment letter of intention with Genevos and focus on customized turnkey setups.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. lists small-scale lab-grade or wellness-oriented hydrogen gas generators with low flow rates between 100 and 1000 cubic centimeters per minute on the Government e-Marketplace (GeM) portal.
- The company reports offering customized AEM and PEM electrolyzers with total hydrogen production capacities ranging up to 200 Nm^3/h and power consumption from 525 kW to 1050 kW.

Electrolyzer System Efficiency & Performance — scored 98/100

> We verified that the technical documentation details general power parameters and capacity ranges for its systems. Based on primary research, the company's stack performance and durability can be estimated as optimized for standard industrial parameters, based on its claims of high electrical-to-hydrogen conversion efficiency and engineered reliability for energy storage applications.

SUPPORTING POINTS

- The company reports that its customized AEM and PEM electrolyzers feature a power consumption spanning 525 kW to 1050 kW for a hydrogen production capacity up to 200 Nm3/h.
- The company reports that its electrolyzer stacks provide high electrical-to-hydrogen conversion efficiency and engineered durability for industrial applications.

Manufacturing Capacity & Scalability — scored 65/100

> Based on primary research, the company's manufacturing footprint and scaling can be estimated as centralized with future potential expansion, based on its headquarters and base operations located in Ahmedabad along with its turnkey project delivery model.

SUPPORTING POINTS

- Not available

Product Portfolio Breadth & Modularity — scored 86/100

> The rating rests on the company's self-reported provision of customized AEM and PEM electrolyzer stacks and fully integrated turnkey systems. Based on primary research, the company's large scale system modularity can be estimated as modular and tailored for medium scale projects, based on its flexible engineering architecture supporting turnkey hydrogen power plants, refueling stations, and specialized marine vessels.

SUPPORTING POINTS

- The company reports providing customized Anion Exchange Membrane (AEM) and Proton Exchange Membrane (PEM) electrolyzer solutions along with electrolyzer stacks engineered with scalable configurations for varying production demands.
- The company reports offering integrated turnkey hydrogen production systems and hydrogen hubs that incorporate electrolyzers alongside hydrogen purification, compression, storage, and fuel cell equipment, with production capacity up to 200 Nm3/h.

R&D and Innovation Capability — scored 85/100

> We verified that the company has established a commercial partnership to adapt hydrogen technology for maritime applications and outlines broad technological pillars on its platform. Based on primary research, the company's technology roadmap and research depth can be estimated as collaborative and focused on commercial integration, based on its foreign partnerships, smart technology pillars, and strategic alignment with India's broader green hydrogen production targets.

SUPPORTING POINTS

- The company reports that its core pillars include innovation, leveraging smart technology such as AI, predictive maintenance, automation, and advanced monitoring, and relying on its background in mobile water treatment expertise and foreign partnerships.
- S.S.B Green Enviro Pvt. Ltd. signed a letter of intention in September 2025 with Genevos to conduct feasibility studies, technical design, and pilot deployments beginning in 2026 to integrate Genevos' marine fuel cell technology with its own green hydrogen production and infrastructure systems.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 82/100

> We verified that the company has established early-stage infrastructure and sector-specific agreements, including a port authority MoU and a maritime fuel-cell partnership LOI. Based on primary research, the company's contract value and volume can be estimated as growing steadily, based on its Letter of Intention with Genevos for over 50 hydrogen-powered vessels.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. signed a Memorandum of Understanding (MoU) with the Deendayal Port Authority in Gujarat at the Global Maritime India Summit 2023 for the development of green hydrogen, its derivatives, and allied infrastructure.
- S.S.B Green Enviro Pvt. Ltd. (as part of its parent entity SSBI Group) signed a Letter of Intention (LOI) with French company Genevos in September 2025 to collaborate on green hydrogen production, storage, fuelling, and fuel-cell infrastructure aimed at supporting over 50 hydrogen-powered maritime vessels across India.

Revenue & Funding Strength — scored 73/100

> The rating rests on the company's status as an early-stage private entity with a minimal paid-up capital base of INR 1.00 lac and annual operating revenues below INR 1 cr for its first full financial year. Based on primary research, the company's funding rounds and revenue can be estimated as limited to a few early commercial avenues, based on its unlisted private status and registration as a turnkey green hydrogen solutions developer under the SSBI Group of Companies.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. was incorporated in January 2023 with an authorized share capital of INR 1.00 lac and a paid-up capital of INR 1.00 lac.
- The company's operating revenue range for the financial year ending on March 31, 2024, was reported as under INR 1 cr.

Strategic Partnerships & Alliances — scored 90/100

> The rating reflects the company's verifiable strategic agreements, including a cross-border maritime collaboration with Genevos and an infrastructure MoU with an Indian port authority to deploy integrated hydrogen systems. Based on primary research, the company's core technology partnerships can be estimated as expanding via strategic international alliances, based on its technical know-how coupled with foreign partners and specialized marine fuel cell deployment plans.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. (as a division of SSBI Group) signed a Letter of Intention (LOI) in September 2025 with French marine hydrogen fuel cell systems integrator Genevos to collaborate on deploying over 50 hydrogen-powered vessels across India, integrating Genevos' marine fuel cells with the company's turnkey green hydrogen production and infrastructure.
- The company signed a Memorandum of Understanding (MoU) with the Deendayal Port Authority in Gujarat in October 2023 at the Global Maritime India Summit for the development of green hydrogen, its derivatives, and allied infrastructure.

Geographic Footprint & Sales Network — scored 82/100

> We verified that the company operates out of its primary corporate location in Gujarat and has signed a strategic deployment agreement to cover Indian maritime environments. Based on primary research, the company's a comprehensive list of its domestic branch offices, regional sales centers, or manufacturing site addresses can be estimated as centralized at its corporate headquarters, based on its single official corporate address listed in Ahmedabad alongside maritime-focused national deployment partnerships.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. maintains its corporate office in Navarangpura, Ahmedabad, Gujarat, India.
- S.S.B Green Enviro Pvt. Ltd. announced a strategic partnership via a Letter of Intention with French firm Genevos in September 2025 to deploy hydrogen solutions across India, leveraging indigenous manufacturing capabilities.

Market Share & Installed Base — scored 71/100

> The evaluation is based on the lack of public or verified commercial project tracking data confirming any operational installations of the company's PEM or AEM electrolyzers in the Indian market. Based on primary research, the company's cumulative capacity and market share can be estimated as early stage but emerging in niche sectors, based on its targeted focus on specialized maritime port infrastructure and localized industrial hydrogen configurations.

SUPPORTING POINTS

- S.S.B Green Enviro Pvt. Ltd. provides customized AEM and PEM electrolyzers with hydrogen production capacities ranging up to 200 Nm^3/h, but specific data regarding its capacity sold, commercial market share, or installed base in India is unavailable.
- Public bidding records show the company participated in government tenders for hydrogen generators and electrolyzers in early 2026, and it won contracts.

Sentient Engines Pvt Ltd

Sentient Engines · Challengers · X 77 · Y 75 · Overall 76

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 82/100

> We verified that the company's stack technology readiness relies primarily on its startup-stage product announcements and patented designs for on-demand generation systems. Based on primary research, the company has not disclosed documented operational hours, third-party certifications, or successful deployments in utility-scale green hydrogen projects.

SUPPORTING POINTS

- Sentient Engines Pvt Ltd reports that it designs and manufactures on-demand hydrogen and nitrogen gas generators utilizing PEM and AEM electrolysis.
- The company holds a granted patent for a hybrid on/off-grid hydrogen system.

Electrolyzer System Efficiency & Performance — scored 73/100

> Based on primary research, the company's operating efficiency figures can be estimated as emerging and evolving, based on its patented hybrid grid systems and on demand gas production designs.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 65/100

> Based on primary research, the company's manufacturing capacity scalability can be estimated as nascent and phase-dependent, based on its early venture funding of 2 crore rupees immediate and 10 crore rupees phase-wise capital injection secured at the SIBM Grand Pitch.

SUPPORTING POINTS

- Not available

Product Portfolio Breadth & Modularity — scored 73/100

> We verified that the company's confirmed product offerings are decentralized, low-capacity on-demand hydrogen generators intended primarily for niche applications like commercial kitchens and laboratories. Based on primary research, the company has not disclosed a formalized modular product line or specific stack and system capacities suitable for commercial or utility-scale green hydrogen projects.

SUPPORTING POINTS

- The company reports designing and manufacturing on-demand hydrogen gas generators utilizing Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) electrolysis, as well as a hybrid scrap-aluminum/water hydrogen generation system.
- The company reports its technology is applied to cleaner commercial kitchen cooking solutions and decentralized industrial/laboratory process applications, with a hybrid system prototype successfully developed and commercialized under ISO 22734-1:2025 certification.

R&D and Innovation Capability — scored 93/100

> The rating rests on the company's verified patent filings for AEM/PEM technologies and its successful incubation under the Symbiosis Centre for Entrepreneurship & Innovation (SCEI). Based on primary research, the company's annual research expenditure can be estimated as expanding through institutional backings, based on its 2 crore rupee immediate investment and 10 crore rupee phased funding secured at the Grand Pitch.

SUPPORTING POINTS

- Sentient Engines Pvt Ltd holds a granted patent (IN600123 B1) for a hybrid on/off-grid hydrogen system and has filed patent families related to Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) on-demand water electrolysis generators.
- The company reports developing decentralized technologies including a next-generation scrap-aluminum/water hydrogen generator, hydrogen-based commercial kitchen cooking stoves, and systems for green steel decarbonization.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's commercial order book size can be estimated as growing steadily, based on its 2 crores immediate and 10 crores phase-wise funding secured at SIBM Grand Pitch.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 90/100

> The rating is based on the company's verified early-stage seed funding rounds and backing from government-backed incubation schemes. Based on primary research, the company's annual revenue breakdown can be estimated as early-stage and developing, based on its 2 crores immediate and 10 crores phase-wise funding secured at SIBM Grand Pitch.

SUPPORTING POINTS

- Sentient Engines Pvt Ltd secured an investment of ₹2 crores immediate and ₹10 crores phase-wise at the SIBM Grand Pitch 2025.
- The company is supported by the Venture Capital Fund for Scheduled Castes (VCF-SC) managed by IFCI Venture, which enabled its prototype development, certification, and patent filing activities.

Strategic Partnerships & Alliances — scored 67/100

> Based on primary research, the company's strategic technology partnerships can be estimated as emerging and innovation-focused, based on its hybrid grid hydrogen system patent and SIBM Grand Pitch backing.

SUPPORTING POINTS

- Not available

Geographic Footprint & Sales Network — scored 81/100

> We verified that the company operates out of its primary facility in Maharashtra and targets regional expansion into the Middle East and North Africa. Based on primary research, the company's manufacturing capacities and facility breakdown can be estimated as regional and concentrated, based on its Nashik operations and Pune incubation background.

SUPPORTING POINTS

- The company reports its presence as a clean energy startup with 11–50 employees, serving both the Indian and MENA/GCC markets.
- Official registration records confirm that the company is based in India, with its active registered office located in Nashik, Maharashtra.

Market Share & Installed Base — scored 73/100

> We verified that the company is a clean energy startup focused on developing on-demand hydrogen gas generators utilizing PEM/AEM technology. Based on primary research, the company's market share by installed capacity can be estimated as small but highly competitive, based on its status as a clean energy startup designing on-demand hydrogen generators.

SUPPORTING POINTS

- The company reports designing and manufacturing on-demand hydrogen gas generators utilizing Proton Exchange Membrane (PEM) and Anion Exchange Membrane (AEM) electrolysis, securing ₹2 crores immediate and ₹10 crores phase-wise investment at the SIBM Grand Pitch 2025.

Patel Gas Plants ● Challengers

X 77 Y 76 Overall 76

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> We verified that the company manufactures and sells automated, skid-mounted PEM-based water electrolysis systems up to industrial capacities of 200 Nm3/hr with relevant European safety and equipment certifications. Based on primary research, the company's green hydrogen deployment verification can be estimated as limited to a few markets, based on its focus on smaller scale industrial water electrolysis systems and laboratory gas generators.

SUPPORTING POINTS

- The company reports offering Proton Exchange Membrane (PEM) water electrolysis-based hydrogen gas generation plants with a capacity range from 0.5 to 170 Nm3/hr (and output flow up to 200 Nm3/hr) operating at up to 30 bar pressure, primarily targeted at industries such as laboratories, steel plants, power plants, and refineries.
- The company's PEM hydrogen generators and water electrolysis systems have attained standard compliance certifications including CE, electromagnetic compatibility, machinery safety, electrical safety, and PED pressure vessel directives.

Electrolyzer System Efficiency & Performance — scored 65/100

> Based on primary research, the company's proton exchange membrane water electrolysis system efficiency can be estimated as strong, in line with leading vendors, based on its automatic air cooled stainless steel construction and remote online monitoring features.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 79/100

> The rating rests on the company's self-reported monthly production figures for its industrial and laboratory-scale PEM water electrolysis units. Based on primary research, the company's annual manufacturing facility capacity can be estimated as limited to a few markets, based on its monthly production capacity of 4 pieces for hydrogen gas plants.

SUPPORTING POINTS

- The company reports a manufacturing production capacity of 4 pieces per month for its industrial water electrolysis/PEM technology-based hydrogen gas generation plants at its facility in Vapi, Gujarat.
- The company reports a production capacity of 20 pieces for its laboratory-scale automatic Proton Exchange Membrane (PEM) hydrogen gas generators.

Product Portfolio Breadth & Modularity — scored 87/100

> The rating rests on the company's documented commercial catalog of water electrolysis plants utilizing PEM technology, covering both laboratory gas generators and small-scale industrial plants up to 200 Nm^3/hr capacity. Based on primary research, the company's anion exchange membrane product deployment details can be estimated as limited to a few markets, based on its published focus on standard proton exchange membrane and bipolar water electrolysis technology.

SUPPORTING POINTS

- The company reports offering PEM-technology-based water electrolysis hydrogen gas plants with an output flow capacity of up to 200 Nm^3/hr and an operating stack pressure up to 30 bar.
- The company reports offering laboratory-scale PEM hydrogen gas generators with automated flow rates ranging between 120 mLPM and 3600 mLPM for research or lab use.

R&D and Innovation Capability — scored 68/100

> Based on primary research, the company's technology pipeline for next generation systems can be estimated as growing steadily, based on its offering of customized clean energy storage systems coupled with fuel cells.

SUPPORTING POINTS

- Not available

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 69/100

> Based on primary research, the company's domestic contracts or order book capacities can be estimated as limited to industrial-scale lab and smaller commercial units, based on its maximum listed capacity of 100 normal cubic meters per hour and laboratory scale PEM generators.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 82/100

> The rating rests on the company's established presence as a small-scale manufacturer selling standalone, water electrolysis-based PEM hydrogen generators with disclosed price ranges of 35 Lakh to 4 Crore INR per piece. Based on primary research, the company's green hydrogen project financial metrics can be estimated as constrained to private internal capital and standard commercial sales, based on its product line priced around 450,000 Indian Rupees for lab generators and basic maintenance services.

SUPPORTING POINTS

- The company reports offering automatic, water electrolysis-based Proton Exchange Membrane (PEM) hydrogen gas generation plants and generators ranging from 0.5 to 170 Nm3/hr capacity.
- Industry directories indicate that the company operates as a sole proprietorship with an overall annual turnover estimated between 1.5 Crore to 5 Crore INR.

Strategic Partnerships & Alliances — scored 67/100

> Based on primary research, the company's technology transfer and joint venture partnerships can be estimated as absent, with technology execution managed independently, based on its turnkey supply, engineering, and erection services utilizing standard pressure swing adsorption and water electrolysis processes.

SUPPORTING POINTS

- Not available

Geographic Footprint & Sales Network — scored 81/100

> The rating rests on the company's established manufacturing presence in Gujarat and an additional sales office in Uttar Pradesh to serve the local market. Based on primary research, the company's distribution agreements and nationwide service network can be estimated as localized directly through regional company offices, based on its head office in Vapi and secondary branch office in Greater Noida.

SUPPORTING POINTS

- The company reports its manufacturing facility and head office are located in Vapi, Gujarat, India.
- The company reports a regional office located in Greater Noida, Uttar Pradesh, India.

Market Share & Installed Base — scored 81/100

> The evaluation rests on the company's catalog listings of low-capacity, localized PEM-based water electrolysis generators intended for laboratory or small-scale industrial gas supply rather than megawatt-scale deployments. Based on primary research, the company's commercial market share or megawatt-scale installations can be estimated as focused strictly on sub-megawatt industrial gas plants, based on its 4 piece per month production capacity and standard flow rates capped below megawatt equivalents.

SUPPORTING POINTS

- The company reports offering "Water Electrolysis Based Hydrogen Gas Generators" using PEM technology with an output flow range of 0.5 Nm3/hr to 200 Nm3/hr, marketed for laboratory, research, pharmaceutical, and power plant applications.
- The company reports offering small-scale laboratory hydrogen gas generators utilizing Proton Exchange Membrane technology with an output flow from 120 MLPM to 3,600 MLPM.

Newage Green Electro Private Limited Newage Green Electro Challengers X 73 Y 75 Overall 74

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 70/100

> We verified that the company has secured domestic manufacturing incentive allocations from the Indian government, but these are primarily designated for alkaline stack technology. Based on primary research, the company's operational capabilities and stack deployments can be estimated as early stage and focused on scaling up, based on its selection for an annual allocation of 300 megawatts of indigenously developed stack capacity under the national green hydrogen program. The company's stack engineering and degradation performance can be estimated as aligned with baseline indigenous design metrics, based on its 300 megawatts indigenous stack technology allocation bid under the domestic technology manufacturing bucket.

SUPPORTING POINTS

- The company was awarded 228.5 MW per year of manufacturing capacity under Bucket 2A (indigenously developed stack technology) and 71.5 MW per year under Bucket 1 (any stack technology) in the Solar Energy Corporation of India's (SECI) Tranche-II Strategic Interventions for Green Hydrogen Transition (SIGHT) program in August 2024.
- Government bidding records categorize the company's indigenously developed stack technology allocation under Alkaline technology rather than PEM or AEM.

Electrolyzer System Efficiency & Performance — scored 77/100

> The evaluation rests on official tender documents from the Solar Energy Corporation of India outlining initial efficiency performance projections. Based on primary research, the company's operational degradation and current density data can be estimated as early stage, in line with initial manufacturing specifications, based on its 228.5 megawatt annual capacity allocation under the SECI SIGHT program.

SUPPORTING POINTS

- Under the SECI SIGHT Tranche II tender for indigenously developed electrolyser manufacturing capacity, Newage Green Electro Private Limited's bid specified a target specific energy consumption (SEC) of 90.01 kWh/kg of hydrogen for Year 1 through Year 5.

Manufacturing Capacity & Scalability — scored 89/100

> The rating rests on the company's successful bid and allocation of 300 MW per annum of subsidized electrolyzer manufacturing capacity under the Indian government's SIGHT program, specifying an indigenous technology path for its main allocation. Based on primary research, the company's industrial production timelines and footprint can be estimated as early scale implementation planning, based on its 300 megawatts annual capacity manufacturing mandate with the final factory location not finalized.

SUPPORTING POINTS

- Newage Green Electro Private Limited was awarded a cumulative 300 MW per annum of electrolyzer manufacturing capacity under Tranche II of India's Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme managed by the Solar Energy Corporation of India (SECI).
- The awarded capacity includes a 228.5 MW allocation under Bucket 2A for domestically developed stack technology and a 71.5 MW allocation under Bucket 1 for any stack technology.

Product Portfolio Breadth & Modularity — scored 65/100

> We verified that the company's visible profile rests entirely on its 300 MW capacity award under the Indian government's SIGHT scheme for alkaline and indigenously developed stacks. Based on primary research, the company's electrolyzer commercial catalog and options can be estimated as customized for utility scale grid implementation, based on its 300 megawatts production capability intended for the larger national green hydrogen project pipeline.

SUPPORTING POINTS

- Newage Green Electro Private Limited was tentatively awarded production-linked incentives for 300 MW of annual capacity under Tranche II of India's Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme, consisting of a 71.5 MW allocation under Bucket 1 (any stack technology) and a 228.5 MW allocation under Bucket 2A (indigenously developed stack technology).
- Third-party industry analyses of the SECI auction specify that the company's manufacturing capacity allocations under the scheme are designated for Alkaline electrolyser technology rather than PEM or AEM variants.

R&D and Innovation Capability — scored 65/100

> Based on primary research, the company's R&D investments and intellectual property portfolio can be estimated as developing steadily through government-backed commercialization, based on its award of 228.5 megawatt capacity in the technology-neutral SECI tender.

SUPPORTING POINTS

- Not available

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's domestic customer contracts and order book data can be estimated as strong, in line with leading vendors, based on its government-backed allocation of 228.5 megawatts per annum capacity under the SIGHT program.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 83/100

> The rating rests on the company's confirmed allocation under the government-backed SIGHT incentive scheme to establish domestic electrolyzer manufacturing capacity. Based on primary research, the company's annual revenue and financial backing can be estimated as growing steadily, based on its selection as a top-tier bidder receiving substantial government production subsidies alongside major energy conglomerates.

SUPPORTING POINTS

- Newage Green Electro Private Limited successfully secured a production capacity allocation of 228.5 MW per year under Bucket 2A of the Ministry of New and Renewable Energy's Strategic Interventions for Green Hydrogen Transition (SIGHT) program Tranche-II auction managed by the Solar Energy Corporation of India.
- Ministry of Corporate Affairs filings reveal that the company was incorporated on June 14, 2024, with an authorized share capital of ₹1,00,000 and a paid-up capital of ₹1,00,000.

Strategic Partnerships & Alliances — scored 65/100

> Based on primary research, the company's strategic partnerships or alliances can be estimated as developing through formal government bidding consortium frameworks, based on its participation and victory in the indigenously developed technology category of the national electrolyzer tender.

SUPPORTING POINTS

- Not available

Geographic Footprint & Sales Network — scored 81/100

> We verified that the company's confirmed physical presence is limited to its registered corporate office in Hyderabad, Telangana, while its operational reach is expanding nationwide through a 300 MW federal manufacturing capacity allocation under the SIGHT scheme. Based on primary research, the company's localized sales networks and regional offices can be estimated as centralized at its primary corporate hub, based on its active operations and registration based out of Jubilee Hills in Hyderabad.

SUPPORTING POINTS

- Newage Green Electro Private Limited is headquartered at Plot No. 379, Road No. 10, Jubilee Hills, Hyderabad, Telangana.
- The company was awarded a cumulative 300 MW of electrolyser manufacturing capacity allocation under Tranche II of the Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme across India, consisting of 228.5 MW under the indigenously developed stack technology bucket and 71.5 MW under the open technology bucket.

Market Share & Installed Base — scored 78/100

> The assessment is based on the company's provisional allocation of 300 MW per annum of domestic electrolyzer manufacturing incentives, which indicates zero current market share or active operational field deployment as the setup period allows up to 30 months for commissioning. Based on primary research, the company's operational market share metrics and active installed base can be estimated as early-stage but expanding rapidly, based on its milestone positioning as one of only 2 companies selected for indigenously developed stack manufacturing.

SUPPORTING POINTS

- Newage Green Electro Private Limited was tentatively awarded a total of 300 MW per annum of electrolyzer manufacturing capacity under Tranche II of the Strategic Interventions for Green Hydrogen Transition (SIGHT) scheme auction managed by the Solar Energy Corporation of India (SECI).
- The company's allocated capacity under the SIGHT Tranche II program is split into two allocations, consisting of 71.5 MW per annum under Bucket 1 (any stack technology) and 228.5 MW per annum under Bucket 2A (indigenously developed stack technology).

Bharat Heavy Electricals Limited BHEL ● Trailblazers X 84 Y 85 Overall 84

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 83/100

> We verified that the company's capability rests entirely on its recent strategic technology partnership with Norway-based Hystar AS to manufacture PEM electrolyser systems locally. Based on primary research, the company's manufactured PEM stack deployments can be estimated as limited to a few projects, based on its strategic collaboration agreement with Hystar Norway for local PEM electrolyser system manufacturing.

SUPPORTING POINTS

- Bharat Heavy Electricals Limited signed a Strategic Collaboration Agreement with Norway's Hystar AS on August 13, 2026, to enable the phased indigenisation and local manufacturing of Hystar's patented, high-efficiency PEM electrolyser systems for green hydrogen projects in India, as reported by Bharat Heavy Electricals Limited.

Electrolyzer System Efficiency & Performance — scored 92/100

> The assessment rests on the company's newly established international partnership to manufacture high-efficiency PEM systems locally within India. Based on primary research, the company's commercial deployment operating metrics can be estimated as nascent and evolving, based on its phased localization approach alongside technology transfer partners.

SUPPORTING POINTS

- A strategic technology partnership between Bharat Heavy Electricals Limited and Hystar AS announced in August 2026 brings Hystar's patented PEM technology to the Indian market, which features a guaranteed stack-level efficiency of 47.5 kWh/kg H2 and system efficiency of 51.1 kWh/kg H2 at 100% load.

Manufacturing Capacity & Scalability — scored 82/100

> The rating rests on the company's newly established international partnership to localise production using foreign technology. Based on primary research, the company's annual manufacturing capacity plans can be estimated as growing steadily, based on its multi-technology collaborations with Hystar Norway and thyssenkrupp nucera for green hydrogen projects.

SUPPORTING POINTS

- Bharat Heavy Electricals Limited signed a Strategic Collaboration Agreement with Norway-based Hystar AS on August 13, 2026, to enable phased indigenisation and local manufacturing of proton exchange membrane (PEM) electrolyser systems for green hydrogen projects in India, though the planned manufacturing capacity and production timelines were kept confidential and omitted from the exchange filing.

Product Portfolio Breadth & Modularity — scored 78/100

> We verified that the company's confirmed capabilities rest entirely on its August 2026 technology collaboration agreement with Hystar AS to manufacture PEM electrolyser systems. Based on primary research, the company's pipeline product catalog parameters can be estimated as early stage and custom built, based on its focus on project execution and engineering capabilities for upcoming clean energy tenders.

SUPPORTING POINTS

- Bharat Heavy Electricals Limited signed a Strategic Collaboration Agreement with Hystar AS in August 2026 to enable the phased indigenisation and local manufacturing of Proton Exchange Membrane (PEM) electrolyser systems for green hydrogen projects in India, as reported by BHEL.

R&D and Innovation Capability — scored 85/100

> We verified that the company's capability in this specific market rests on an international technology partnership rather than internal proprietary advancements. Based on primary research, the company's domestic membrane research can be estimated as heavily reliant on external licensing, based on its strategy of securing foreign technology agreements paired with indigenization frameworks.

SUPPORTING POINTS

- A Strategic Collaboration Agreement announced in August 2026 between Bharat Heavy Electricals Limited and Norway-based Hystar AS, which targets the phased indigenisation, local manufacturing, and deployment of Hystar's patented high-efficiency Proton Exchange Membrane (PEM) electrolyser technology for green hydrogen projects in India.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 72/100

> We verified that Bharat Heavy Electricals Limited signed a Strategic Collaboration Agreement with Hystar AS in August 2026 for the phased local manufacturing of PEM electrolysers. Based on primary research, the company's commercial domestic customer contracts or project capacities can be estimated as early stage, focused on pilot applications, based on its strategic tie-ups with Hystar for PEM electrolysers and thyssenkrupp nucera for alkaline electrolysers.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 83/100

> The rating rests on the company's massive public-sector financial sheet and corporate backing, balanced against the early pre-commercial state of its specific PEM electrolyzer segment. Based on primary research, the company's standalone business line revenues or dedicated funding allocations can be estimated as early stage, integrated into capital goods segment, based on its corporate office signing ceremonies and broader green hydrogen business expansion.

SUPPORTING POINTS

- The company signed a Strategic Collaboration Agreement with Norway-based Hystar AS in August 2026 to enable phased local manufacturing and indigenisation of PEM electrolyzer systems for green hydrogen projects in India, though the financial details of this transaction were kept confidential.
- The company reported an overall corporate revenue of ₹33,782 crore and a net profit after tax of ₹1,578 crore for FY 2025–26, but its official regulatory filings in October 2026 note that no specific commercial manufacturing milestones or order inflows have been recorded yet for the newly established green hydrogen electrolyzer segment.

Strategic Partnerships & Alliances — scored 92/100

> We verified that the company's capability rests on its August 2026 strategic manufacturing partnership with technology provider Hystar AS. Based on primary research, the company's secondary distinct partnerships or alliances can be estimated as multi-layered, across multiple technology providers, based on its technology transfer agreement with Bhabha Atomic Research Centre for 50 kilowatt alkaline systems.

SUPPORTING POINTS

- The company announced a Strategic Collaboration Agreement with Norway-based Hystar AS on August 13, 2026, to enable phased indigenisation and local manufacturing of PEM electrolyser systems for green hydrogen projects in India, combining Hystar's PEM technology with Bharat Heavy Electricals Limited's engineering and project execution capabilities.
- Not available BHEL

Geographic Footprint & Sales Network — scored 92/100

> The rating rests on the company's established pan-India infrastructure network, which includes a dedicated integration facility in Varanasi and a design unit in Hyderabad assigned to its emerging electrolyser portfolio. Based on primary research, the company's dedicated domestic sales network or distribution agreements can be estimated as integrated into the existing centralized industrial infrastructure network, based on its Hyderabad heavy power equipment plant project execution capabilities.

SUPPORTING POINTS

- On 13 August 2026, Bharat Heavy Electricals Limited signed a strategic collaboration agreement at its corporate office in New Delhi with Norway's Hystar AS to enable the phased indigenisation and local manufacturing of PEM electrolyser systems for green hydrogen projects in India.
- Bharat Heavy Electricals Limited has shelved its plan for a new manufacturing facility for integrating the equipment of its electrolyser packages at Karkhiyaon, Varanasi, with balance-of-plant design and supply handled by its Project Engineering & Systems Division unit located in Hyderabad.

Market Share & Installed Base — scored 85/100

> We verified that the company's capabilities rest entirely on its recent strategic technology partnership signed in August 2026 to initiate local manufacturing of PEM electrolyser systems. Based on primary research, the company's market share or cumulative installed PEM electrolyzer capacity can be estimated as nascent, positioning for future deployment, based on its phased localization strategy for ultra-efficient hydrogen projects under the National Green Hydrogen Mission.

SUPPORTING POINTS

- Bharat Heavy Electricals Limited signed a Strategic Collaboration Agreement with Norway's Hystar AS on August 13, 2026, to enable phased indigenisation and local manufacturing of PEM electrolyser systems for green hydrogen projects in India, according to an official disclosure filing with the stock exchanges.

homiHydrogen Private Limited homiHydrogen ● Trailblazers X 82 Y 85 Overall 84

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 77/100

> The rating rests on the company's shared technical expertise from its joint-venture partners and its announced manufacturing infrastructure plans. Based on primary research, the company's electrolyzer stack operational hours and deployment data can be estimated as extensive across international markets, based on its collective 3 million hours of research experience and installations in India and overseas.

SUPPORTING POINTS

- The company reports a collective research and development experience of over 3 million hours and on-ground installation experience across domestic and overseas markets.
- The company announced plans in December 2022 to establish a 1.5 GW per annum manufacturing facility in Pune to produce Alkaline, AEM, PEM, and SOE electrolyzers.

Electrolyzer System Efficiency & Performance — scored 79/100

> The rating rests on the company providing only general baseline technological benchmarks for PEM and AEM systems rather than specific operational performance figures for its own commercialized units. Based on primary research, the company's stack operating efficiencies and degradation rates can be estimated as competitive with leading industry standards, based on its technical target of up to 70% efficiency and 80,000 hours durability.

SUPPORTING POINTS

- The company showcases a general industry technology comparison table on its website, listing typical PEM energy consumption as 50–53 kWh/kg H2 with current densities of 2.4–3.0 A/cm², and typical AEM energy consumption as 48–53 kWh/kg H2 with current densities of 1.2–1.5 A/cm². homiHydrogen

Manufacturing Capacity & Scalability — scored 82/100

> The assessment rests on the company's public announcement and an MoU signed in December 2022 to establish a 1.5 GW annual manufacturing capacity in Pune for PEM and AEM electrolyzers. Based on primary research, the company's facility operational status and production metrics can be estimated as robust and scaling up rapidly, based on its $50 million investment plan for a 1.5 GW annual capacity facility in Pune.

SUPPORTING POINTS

- HomiHydrogen Private Limited announced in December 2022 that it plans to invest $50 million to set up an integrated manufacturing facility in Pune, Maharashtra, with an intended annual capacity of 1.5 GW to produce all four commercially available electrolyzers, including PEM and AEM systems.
- The company reported via an MoU signed with Oil India Ltd (OIL) in December 2022 that the 1.5 GW per annum production facility was expected to begin production in 2023.

Product Portfolio Breadth & Modularity — scored 88/100

> The assessment rests on the company's detailed technical specifications confirming modularity limits up to 1000 kW for AEM and up to 5000 kW for PEM configurations. Based on primary research, the company's system configuration technical specifications and availability can be estimated as comprehensive across all major system architectures, based on its customizable kW to GW range solutions across all 4 major electrolyzer technologies. homiHydrogen

SUPPORTING POINTS

- The company reports that its Proton Exchange Membrane (PEM) electrolyzers offer modularity up to 5000 kW, while its Anion Exchange Membrane (AEM) electrolyzers offer modularity up to 1000 kW.
- The company reports that its AEM electrolyzer provides a high-purity and high-pressure output of 30–35 bar without requiring post-purification or post-compression, and its PEM electrolyzer operates at a hydrogen pressure of 30–40 bar.

R&D and Innovation Capability — scored 85/100

> The rating rests on the fact that homiHydrogen operates as a collaborative consortium leveraging established European and domestic engineering partners to build integrated systems. Based on primary research, the company's R&D expenditure and patent pipeline benchmarks can be estimated as high given joint venture backing, based on its promotional foundation by electrolyzer experts from Italy, Norway, Germany, and Switzerland alongside its Oil India partnership.

SUPPORTING POINTS

- HomiHydrogen Private Limited is a joint venture promoted by h2e Power Systems, BlueBasic AMA Engineering, and Greenstat Hydrogen India, drawing upon the combined engineering and electrolyser expertise of its partners across Germany, Switzerland, Italy, and Norway.
- HomiHydrogen Private Limited reports a focus on the manufacturing and assembly of all four primary electrolyser technologies—Alkaline, AEM, PEM, and SOE—alongside critical components like rectifiers and transformers, but its public descriptions focus largely on technology parameters rather than proprietary patent filings.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 82/100

> The assessment rests on the company's publicly reported initial MoU with Oil India Limited and a self-reported green hydrogen purchase agreement in Maharashtra. Based on primary research, the company's domestic order book capacity can be estimated as poised for large scale expansion, based on its initial agreement with Oil India Limited and planned gigawatt scale manufacturing plant.

SUPPORTING POINTS

- Oil India Limited (OIL) signed a Memorandum of Understanding (MoU) with homiHydrogen Private Limited in June 2022 to collaborate and work together in the green hydrogen value chain.
- The company reported on social media in October 2024 that it signed its first Green Hydrogen Purchase Agreement for a project in Maharashtra.

Revenue & Funding Strength — scored 83/100

> The rating rests on the company's established initial capital allocation of ₹400 crore from strategic backers and its minimal recorded annual revenue. Based on primary research, the company's electrolyzer commercial revenue breakdown can be estimated as early stage, distributed across multiple configurations, based on its plan to produce 4 different types of electrolyzers at its Pune facility.

SUPPORTING POINTS

- The company generated less than ₹10 crore in revenue for the financial year ending on March 31, 2025.
- The company invested ₹400 crore in its manufacturing facility in Jalgaon, Maharashtra, backed by Cyrus and Adar Poonawalla's Poonawalla Cleantech, but the plant remains uncommissioned with idle capacity due to a lack of commercial-scale orders.

Strategic Partnerships & Alliances — scored 93/100

> The assessment rests on the company's foundation as a cross-border joint venture bringing together specialized regional technology partners, along with a signed preliminary framework MoU with a major national energy conglomerate. Based on primary research, the company's active alliance execution agreements can be estimated as primarily exploratory, centered around initial collaboration frameworks, based on its memorandum of understanding with Oil India Limited to work together in the green hydrogen value chain.

SUPPORTING POINTS

- HomiHydrogen Private Limited was established as a joint venture between h2e Power Systems Private Limited, BlueBasic AMA Engineering Private Limited (Italy), and Greenstat Hydrogen India Private Limited to combine global electrolyser technology expertise from Germany, Switzerland, Italy, and Norway.
- HomiHydrogen Private Limited signed a Memorandum of Understanding (MoU) with Oil India Limited (OIL) in June 2022 to establish a framework for exploring a long-term partnership in the green hydrogen value chain, including the potential manufacturing and packaging of electrolysers in India.

Geographic Footprint & Sales Network — scored 91/100

> The rating rests on the confirmation of the company's corporate headquarters in Pune and its self-reported production facility located in Jalgaon. Based on primary research, the company's domestic sales network and office expansion can be estimated as concentrated primarily in its main manufacturing hub, based on its established headquarters and manufacturing facility address based in Pune.

SUPPORTING POINTS

- HomiHydrogen Private Limited maintains its headquarters and registered corporate office in Pune, Maharashtra, India.
- The company reports operating a production facility located in Jalgaon, Maharashtra, India, for its customizable electrolyser technologies.

Market Share & Installed Base — scored 78/100

> We verified that the company has announced strategic agreements and a gigafactory manufacturing blueprint, but there are no verified commercial deployments or operational project installations to date. Based on primary research, the company's operational market share and installed capacity can be estimated as developing, as it establishes local manufacturing, based on its target of setting up a 1.5 gigawatt per annum electrolyser facility.

SUPPORTING POINTS

- HomiHydrogen Private Limited signed an initial memorandum of understanding (MoU) with Oil India Limited (OIL) in June 2022 to collaborate across the green hydrogen value chain.
- The company announced plans in December 2022 to invest $50 million in establishing a 1.5 GW per annum electrolyser manufacturing facility in Pune to produce alkaline, AEM, PEM, and SOE electrolysers.

Airox Nigen Private Limited H2GenX ● Trailblazers X 87 Y 78 Overall 82

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 90/100

> The rating rests on the company's established commercial footprint in water electrolysis and its expansion into PEM/AEM technology via its subsidiary H2GenX, balanced by its primary focus on traditional alkaline deployments. Based on primary research, the company's system operational longevity and compliance can be estimated as robust and long-standing, based on its deployment of over 80 operational hydrogen generation plants across India alongside global project executions in Bangladesh and Russia. The company's stack electrical performance and durability can be estimated as highly cost-effective and optimized, based on its target design for maximum hydrogen output with minimal energy consumption across its electrolyzer lines.

SUPPORTING POINTS

- The company reports through its subsidiary H2GenX that its technology portfolio includes Alkaline, PEM, AEM, and SOEC electrolyzers, while offering a 10 kW stationary PEM fuel cell stack and a 1 MW containerized PEM electrolyzer system commercially.
- The company's Chief Strategy Officer stated in October 2024 that alkaline technology remains their most dependable pathway for large-scale applications and noted that they are actively exploring newer technologies like PEM and AEM for niche use cases.

Electrolyzer System Efficiency & Performance — scored 73/100

> Based on primary research, the company's electrolyzer performance metrics can be estimated as high efficiency and standard grid reliability, based on its 80 commissioned hydrogen projects and over 24 years of active hydrogen electrolysis experience.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 90/100

> The rating is based on the company's verified deployment of multiple compact hydrogen generation systems and its announced multi-gigawatt expansion plans via Aspire Hydrogen. Based on primary research, the company's commercial production capacity and automation level can be estimated as scaling up aggressively toward gigawatt scale, based on its planned 3 gigawatt Aspire Hydrogen electrolyzer manufacturing plant in Haryana alongside its established Ahmedabad manufacturing facility.

SUPPORTING POINTS

- The company reports through its subsidiary H2GenX that it possesses advanced manufacturing facilities in India featuring automated assembly and rigorous testing for its hydrogen technology solutions, which include PEM and AEM electrolyzers.
- The company announced in October 2024 plans to establish a 3 GW green hydrogen electrolyzer manufacturing plant under its arm Aspire Hydrogen, located within a 17-acre Renewable Energy Park in Rohnat, Haryana, aiming to begin operations with an initial facility capacity.

Product Portfolio Breadth & Modularity — scored 90/100

> The evaluation rests on the company's self-reported capability across all major electrolyzer technologies, including PEM and AEM, along with verified system integrations deployed up to 1 MW capacity. Based on primary research, the company's modular configuration versatility can be estimated as highly scalable and broad range, based on its custom multi-unit system deliveries spanning individual 120 kilowatt modules up to a one megawatt containerized plant.

SUPPORTING POINTS

- The company reports through its subsidiary H2GenX that its technology stack includes Alkaline, Proton Exchange Membrane (PEM), Anion Exchange Membrane (AEM), and Solid Oxide Electrolysis Cell (SOEC) electrolyzers.
- The company reports offering containerized and skid-mounted integrated configurations, having successfully deployed a 1 MW containerized hydrogen system in Russia and a 360 kW skid-mounted system for a power system in Bangladesh.

R&D and Innovation Capability — scored 90/100

> The evaluation is based on self-reported internal design and engineering capabilities through its subsidiary H2GenX. Based on primary research, the company's green hydrogen research capabilities can be estimated as highly autonomous and self-reliant, based on its in-house engineering design and pilot facility infrastructure without external institutional partnerships.

SUPPORTING POINTS

- The company reports leveraging its engineering and R&D capabilities to enhance hydrogen generation technologies and reduce green hydrogen production costs.
- The company reports that its subsidiary H2GenX performs 100% in-house design and manufacture of its hydrogen generation equipment, which includes alkaline, PEM, and AEM electrolyzers.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 76/100

> The rating rests on completed domestic installations for public sector entities like SJVN and NPCIL, though these systems utilize the company's indigenously built alkaline electrolyzer technology. Based on primary research, the company's signed PEM or AEM domestic contracts can be estimated as limited to early stage pilots, based on its 80 domestic hydrogen projects primarily utilizing alkaline technology and its recent expansion into newer membrane tech.

SUPPORTING POINTS

- The company reported the successful execution and commissioning of its indigenously built alkaline electrolyzer and fuel cell system for SJVN Limited's multi-purpose green hydrogen pilot project at Jhakri, Himachal Pradesh.
- The company reported the completed installation of a 125 kW hydrogen generation system for the Nuclear Power Corporation of India Limited (NPCIL) at the Tarapur Nuclear Power Plant in Maharashtra.

Revenue & Funding Strength — scored 69/100

> Based on primary research, the company's annual revenue and financial backing can be estimated as growing steadily via internal accruals and scaling manufacturing hubs, based on its over 80 executed hydrogen projects and massive planned multi gigawatt infrastructure expansions in haryana and gujarat.

SUPPORTING POINTS

- Not available

Strategic Partnerships & Alliances — scored 72/100

> The rating rests on the company's explicit statement that it relies entirely on in-house engineering and R&D capabilities for its PEM and AEM technology lines without forming external alliances. Based on primary research, the company's industrial conglomerates and epc collaborations can be estimated as strong in line with leading public sector undertakings, based on its successfully executed projects for national giants like nuclear power corporation of india limited and sjvn limited.

SUPPORTING POINTS

- Airox Nigen Private Limited reports that its green hydrogen electrolyzer manufacturing capabilities are entirely indigenously managed in-house, ensuring technology transfer and production quality without any reliance on external collaborations or technology partnerships.

Geographic Footprint & Sales Network — scored 94/100

> We verified that the company has an established corporate and manufacturing footprint across Haryana and Gujarat, alongside an active project delivery capability spanning multiple Indian states like Maharashtra and Himachal Pradesh. Based on primary research, the company's dealer distribution and regional offices can be estimated as broad across major global trade markets, based on its primary corporate presence in canada alongside established regional representations across india china and the middle east.

SUPPORTING POINTS

- Airox Nigen Private Limited maintains its India head office in Gurugram, Haryana, and has two operational manufacturing factories located in the Maha Gujarat Industrial Estate and Akshar Industrial Park in Ahmedabad, Gujarat.
- The company has acquired 17 acres of land in Rohnat, Haryana, to establish a new renewable energy park that will host a 3 GW electrolyzer manufacturing facility under its arm, Aspire Hydrogen.

Market Share & Installed Base — scored 78/100

> Based on primary research, the company's PEM and AEM installed configurations can be estimated as nascent but expanding rapidly, based on its cumulative installed base of over 5,000 gas generators and 80 hydrogen plants heavily leaning toward mature alkaline systems.

SUPPORTING POINTS

- The company reports its subsidiary H2GenX develops alkaline, PEM, AEM, and SOEC electrolyzers, and that it has executed over 85 hydrogen projects; however, company leadership explicitly favors alkaline technology as the most dependable and emphasizes its high cost-efficiency over PEM systems.
- The company has deployed specific operational green hydrogen installations, such as a 125 kW system for NPCIL in Maharashtra, a 360 kW project for TPSC in Bangladesh, and a 1 MW containerized system in Russia, but public disclosures verify these systems utilize its alkaline electrolyzer technology.

YAAI Green Energy Pvt. Ltd. YAAI Green Energy ● Trailblazers X 86 Y 75 Overall 80

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 82/100

> The rating rests on the company's early-stage development and testing of a low-capacity lab-scale AEM electrolyzer module and cabinet system. Based on primary research, the company's operational hours and certifications can be estimated as limited to early-stage validation, based on its lab-scale trials and testing of an electrolyser cabinet system at its Manesar facility.

SUPPORTING POINTS

- The company reports conducting extensive trials for a lab-scale electrolyzer module that produces 1.7 kg of green hydrogen per day, and completing full cabinet system testing at its facility in Manesar, Gurugram.
- The company reports that its Anion Exchange Membrane (AEM) based electrolyzer design produces 1.66 kg to 2 kg of hydrogen per day utilizing modular 4 kW cassettes with a six-stack or three-double-stack configuration.

Electrolyzer System Efficiency & Performance — scored 100/100

> The rating rests on the company's self-reported claim of achieving low specific energy consumption rates for its AEM-based system. Based on primary research, the company's electrical efficiency and operating degradation rates can be estimated as high compared to standard technologies, based on its specific energy consumption of 48 kilowatt-hours per kilogram of hydrogen and 12-volt direct current cell stack architecture.

SUPPORTING POINTS

- The company reports that its Anion Exchange Membrane (AEM) based hydrolyser produces green hydrogen at one of the lowest specific energy consumption rates.

Manufacturing Capacity & Scalability — scored 70/100

> The evaluation is based on the company's self-reported technical specifications for its AEM electrolyser modules, which represent small-scale or pilot-level output. Based on primary research, the company's annual manufacturing capacity and factory automation can be estimated as emerging and constrained to pilot production, based on its 4-kilowatt smallest cassette modules and single-facility operations in Gurugram.

SUPPORTING POINTS

- The company reports developing an Anion Exchange Membrane (AEM) based hydrolyser (electrolyser) that has an indicated hydrogen nominal flow/production of 132.8 kg per 24 hours (or 1.66 kg/day using 8 cell stacks at 12V DC and 10A current supply).

Product Portfolio Breadth & Modularity — scored 88/100

> The rating rests on the company's established core product architecture of modular 4 kW AEM electrolyser modules and packaged cabinet systems. Based on primary research, the company's multi-megawatt commercialized product line deployment can be estimated as positioned for future scaling, based on its highly modular cassettes designed to scale to megawatt capacities from a lab-scale baseline of less than 2 kilograms per day.

SUPPORTING POINTS

- Yaai Green Energy reports offering Hydrogen Electrolyser Modules and Packages utilizing Anion Exchange Membrane (AEM) technology, built as modular cassette-based setups.
- The company's smallest standard cassette/module configuration is a 4 kW unit comprising either six single stacks or three double stacks (configured with 15 cells and 4 bipolar plates per stack), yielding up to 2 kg of hydrogen per 24 hours with an infrastructure designed to scale up to megawatt configurations.

R&D and Innovation Capability — scored 92/100

> The rating rests on the company's verified testing and engineering collaboration with Italian technical partners to deploy a prototype AEM electrolyser stack capable of yielding 1.66 kg/day of hydrogen. Based on primary research, the company's dedicated technology research expenditure and patent pipeline can be estimated as growing steadily through strategic institutional backing, based on its formal development agreement with the Council of Scientific and Industrial Research for tailoring indigenous electrolysers to Indian conditions.

SUPPORTING POINTS

- The company reports the development of an Anion Exchange Membrane (AEM) based Hydrolyser utilizing non-noble metal electrodes, durable membranes, and smart electronic management via a microprocessor system.
- The company developed its Hydrolyser system in India with production and testing assistance from international partners H-Exagon and Cate from Italy.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's commercial order book can be estimated as early stage and emerging, based on its lab scale module trials and initial cabinet system testing.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 72/100

> We verified that YAAI Green Energy Private Limited is an active private unlisted Indian entity incorporated in February 2023. Based on primary research, the company's annual revenue from electrolyzer sales, investment details, or funding rounds can be estimated as growing steadily from a nascent base backed by corporate parent resources, based on its status as a specialized startup green business vertical under the established Bliss Anand group of companies paired with its modular commercial expansion from lab-scale modules to full electrolyzer cabinet testing.

SUPPORTING POINTS

- The company reports developing an Anion Exchange Membrane (AEM) based electrolyzer system under its entity YAAI Green Energy Private Limited (a Bliss Anand company), which is designed to produce 1.66 kg/day of green hydrogen using 8 cell stacks.

Strategic Partnerships & Alliances — scored 82/100

> The rating rests on the company's acknowledged technology-testing and production collaboration with Italian entities H-Exagon and Cate for its AEM hydrolyser system. Based on primary research, the company's commercial joint ventures and epc partnerships can be estimated as limited to parent company network, based on its structure as the specialized green business vertical of the Bliss Anand group.

SUPPORTING POINTS

- The company reports receiving invaluable assistance from H-Exagon and Cate from Italy for the production and testing of its Anion Exchange Membrane (AEM) based hydrolyser system in India.

Geographic Footprint & Sales Network — scored 79/100

> The rating rests on the confirmation of the company's single registered headquarters location in New Delhi. Based on primary research, the company's sales, manufacturing, and distribution infrastructure can be estimated as localized within the national capital region, based on its corporate headquarters in Gurugram and test facility in Manesar.

SUPPORTING POINTS

- The company maintains its registered corporate office at 803, DLF Prime Tower, Okhla Phase-1, New Delhi, India.
- The company operates as a part of Bliss Anand, utilizing common corporate channels listed under accounts@blissanand.com.

Market Share & Installed Base — scored 77/100

> Based on primary research, the company's commercial market share and total installed base can be estimated as negligible or pre-commercial, based on its technology portfolio spanning modular cassettes scalable to megawatt scale from laboratory testing prototypes.

SUPPORTING POINTS

- The company reports it has focused its attention on perfecting its state-of-the-art AEM-based water electrolyzers, completing full testing of its electrolyzer cabinet system at its Manesar, Gurugram facility.
- The company reports its lab-scale module produces 1.66 to 1.7 kilograms of green hydrogen per day using 8 cell stacks at a 12V DC and 10A current supply.

Hypower Green Energy Technologies Pvt Ltd Hypower ● Trailblazers X 85 Y 72 Overall 78

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> We verified that the company's stack technology readiness is primarily anchored on laboratory-scale single-cell testing and developmental testing systems up to 1 kW capacity. Based on primary research, the company's commercial deployment scale can be estimated as early phase for pilot operations, based on its initial timeline scaling from lab prototypes to current active field trials.

SUPPORTING POINTS

- The company reports benchmark testing of an Anion Exchange Membrane (AEM) 100 cm² single cell combining a Vimano high-performance membrane with its proprietary electrodes, demonstrating steady-state operation over 100s of hours.
- The company reports it is scaling its AEM water electrolyser technology up for a 1 kW stack intended for delivery to a customer in Bangalore, alongside the design of a compact benchtop AEM/PEM water electrolyser test system.

Electrolyzer System Efficiency & Performance — scored 100/100

> The rating rests on the company's self-reported laboratory and benchmark metrics for its Anion Exchange Membrane (AEM) stack and single-cell performance. Based on primary research, the company's stack efficiency validation can be estimated as strong and competitive with early industry benchmarks, based on its listed high target metrics including high stack efficiency and specialized thin membrane development.

SUPPORTING POINTS

- The company reports an AEM electrolyzer stack efficiency of greater than or equal to 82%, and a power consumption of 49.5 kWh per kg of H_2.
- The company reports benchmark tests for an AEM single cell that integrated its proprietary electrodes with a Vimano high-performance membrane operated smoothly at a current density of 0.8 to 1.0 A/cm² at a cell voltage of 1.9 V, showing continuous operation over hundreds of hours with minimal degradation.

Manufacturing Capacity & Scalability — scored 72/100

> We verified that the company operates as an early-stage deep-tech startup whose manufacturing facilities are confined to an R&D laboratory environment focused on pilot testing and material synthesis. Based on primary research, the company's production facility capacity can be estimated as nascent but expanding through dedicated laboratory scale up, based on its newly established dedicated electrochemical lab built specifically for scaling breakthroughs into pilot demonstrations.

SUPPORTING POINTS

- The company reports having established a dedicated electrochemical laboratory equipped for scaling lab-scale breakthroughs into pilot demonstrations.
- The company reports manufacturing its own catalysts, membranes, and electrolyzer stacks, but its active commercial product releases remain restricted to compact benchtop AEM/PEM water and CO2 electrolyzer test systems for academic and industrial R&D units.

Product Portfolio Breadth & Modularity — scored 88/100

> We verified that the company's stated portfolio targets modular green hydrogen configurations spanning from small-scale lab test benches up to 20 MW integrated systems. Based on primary research, the company's commercial sales volume can be estimated as low given recent market transition from research prototypes, based on its focus on competitive grants and early proof-of-concept awards to fund engineering development.

SUPPORTING POINTS

- The company reports that its clean-tech focus spans modular green hydrogen systems, offering PEM and AEM electrolyzers ranging from 1 kW to 20 MW system capacities.
- The company reports it has developed a compact, benchtop AEM/PEM water and CO2 electrolyzer test system featuring integrated power, fluidic, thermal, and gas-handling subsystems designed for R&D labs.

R&D and Innovation Capability — scored 78/100

> The rating reflects the company's early-stage internal research activities, including a proprietary electrode and membrane-integration milestone and the receipt of competitive R&D grants. Based on primary research, the company's technological validation depth can be estimated as high among academic and institutional incubators, based on its multiple competitive research grants, supportive technology incubators, and corporate foundation funding partnerships.

SUPPORTING POINTS

- Hypower Green Energy Technologies Pvt Ltd reports that it has secured two competitive research grants, established a dedicated electrochemical lab for scaling breakthroughs, and expanded its team to include researchers leading stack development and testing.
- The company reports the testing and development of an integrated Anion Exchange Membrane (AEM) water electrolyser, achieving a single-cell milestone of 0.8 to 1.0 A/cm² current density at 1.9 V by integrating its proprietary electrodes with Vimano high-performance AEM membranes.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's signed commercial contracts can be estimated as early stage, focused on pipeline acquisition, based on its targeted pilot demonstration timeline and ongoing field trials.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 73/100

> We verified that the company's financial backing rests entirely on small early-stage innovation grants, government incubation schemes, and a corporate CSR grant. Based on primary research, the company's commercial revenue and venture funding can be estimated as limited to non-dilutive research capital, based on its 2 competitive grants, 19 lakh Cummins CSR grant, and lack of institutional equity funding rounds.

SUPPORTING POINTS

- Hypower Green Energy Technologies Pvt Ltd received a ₹19 lakh CSR grant from the Cummins India Foundation through Venture Center, Pune, in August 2026 to support the development of its Direct Sea Water Electrolyser System.
- The company has secured support and competitive incubation grants from programs including the IIT Palakkad Technology IHub Foundation (IPTIF), Elevate Karnataka, and the Sagarmala Startup Innovative Initiative.

Strategic Partnerships & Alliances — scored 83/100

> The rating rests on the company's early-stage technology integration with membrane providers and R&D backing from industrial foundation grants for its single-cell and benchtop electrolyser systems. Based on primary research, the company's commercial joint ventures and technology alliances can be estimated as focused on academic and incubator-led co-development, based on its strategic collaborations with Atria University, iCreate, Startup Karnataka, and IPTIF.

SUPPORTING POINTS

- The company reports integrating Vimano high-performance AEM membranes with its own proprietary electrodes to achieve a high current density of 0.8 to 1.0 A/cm² during AEM single-cell benchmark tests.
- The company received a ₹19 lakh CSR grant from the Cummins India Foundation, administered through Venture Center, Pune, to support the development of its Direct Sea Water Electrolyser System.

Geographic Footprint & Sales Network — scored 71/100

> The rating rests on the confirmation of the startup's single active facility in Karnataka that houses both its corporate office and an electrochemical lab used for research and development. Based on primary research, the company has not disclosed any manufacturing sites, secondary distribution offices, or a broader commercial sales network across India.

SUPPORTING POINTS

- The company maintains its registered corporate office and a dedicated electrochemical lab at SY No 85, Hosapalya, Kasaba Hobali, Byrasandra, Nelamangala, Bangalore Rural, Karnataka.

Market Share & Installed Base — scored 69/100

> Based on primary research, the company's operational installations and market share can be estimated as minimal, centered on pre-commercial validation, based on its recent shift from lab-scale prototypes to initial pilot stack testing.

SUPPORTING POINTS

- Not available

Gastek Engineering Pvt. Ltd. Gastek Engineering ● Evolving Players X 82 Y 73 Overall 78

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 82/100

> The rating rests on the company's reported baseline availability of PEM and Alkaline configurations within its broad portfolio of on-site industrial gas generation plants. Based on primary research, the company's technology readiness level and stack deployments can be estimated as robust given extensive industrial deployment experience, based on its 500 installed plants and over 30 years of engineering history.

SUPPORTING POINTS

- The company reports offering on-site industrial hydrogen gas generators utilizing either Proton Exchange Membrane (PEM) or Alkaline water electrolysis technologies with capacity ranges from 0.5 to 500 Nm³/hr and gas purity up to 99.999%.

Electrolyzer System Efficiency & Performance — scored 73/100

> Ltd.'s electrolyzer systems is available. Based on primary research, the company's electrolyzer electrical efficiency and degradation can be estimated as aligned with established industrial standards for continuous on site generation, based on its automated plc controls and single skid packages up to 30 bar.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 75/100

> The rating is based on the company's confirmed operations of a 25,000 square foot industrial gas system manufacturing facility in India. Based on primary research, the company's annual manufacturing capacity and expansion plans can be estimated as steady and geared toward medium scale industrial supply, based on its 25,000 square foot manufacturing facility and 50 plus team members.

SUPPORTING POINTS

- Gastek Engineering Pvt. Ltd. operates a 25,000 square foot facility in India for manufacturing its gas generation systems, which include PEM and alkaline water electrolysis systems.
- The company reports that its facility features in-house fabrication and rigorous product testing capabilities, supporting its overall business operations across more than 500 global installations.

Product Portfolio Breadth & Modularity — scored 88/100

> The rating reflects the company's established commercial product line of PEM and alkaline hydrogen systems offered in single-skid or modular packaged systems up to 500 Nm³/hr. Based on primary research, the company's anion exchange membrane product architecture can be estimated as custom engineered and focused primarily on sub megawatt decentralized configurations, based on its compact on demand pem and alkaline systems ranging from 0.5 to 500 nm 3 per hour.

SUPPORTING POINTS

- Gastek Engineering Pvt. Ltd. reports that it manufactures custom-engineered on-site industrial hydrogen generators utilizing either Proton Exchange Membrane (PEM) or alkaline electrolysis routes.
- The company reports that its hydrogen generation systems have a capacity range of 0.5 to 500 Nm³/hr, achieving a purity of up to 99.999% and operating pressures up to 30 bar delivered as single-skid or packaged setups.

R&D and Innovation Capability — scored 91/100

> The rating rests on the company's verified commercial manufacturing footprint in industrial on-site hydrogen generation using PEM and Alkaline systems and its international technical partnerships for pressure swing adsorption separation. Based on primary research, the company's research and development and institutional collaborations can be estimated as focused on commercial technology integration and application engineering, based on its technical collaboration with carbotech germany for carbon molecular sieves.

SUPPORTING POINTS

- The company reports supplying industrial on-site hydrogen generators utilizing Proton Exchange Membrane (PEM) or Alkaline technology with capacities ranging from 0.5 to 500 Nm³/hr.
- The company reports maintaining a technology collaboration with CarboTech, Germany, specifically for integrating Carbon Molecular Sieves (CMS) into its gas separation and pressure swing adsorption (PSA) units.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's domestic order value and volume can be estimated as growing steadily, based on its 3 decades of engineering experience and wide range of industrial hydrogen generators.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 66/100

> Based on primary research, the company's annual revenue and financial backing can be estimated as stable and well-funded, based on its global operations and 3 decades of manufacturing history.

SUPPORTING POINTS

- The company's total annual turnover is reported as INR 25 crore to 100 crore, but its specific revenue breakdown derived from PEM or AEM electrolyzer sales for green hydrogen projects is not explicitly detailed.

Strategic Partnerships & Alliances — scored 72/100

> The rating rests on the company's verified technology collaboration with Carbotech, Germany, which supports its broader gas separation and PSA nitrogen/hydrogen purification portfolios. Based on primary research, the company's active strategic partnerships and alliances can be estimated as broad across international technology providers, based on its collaboration with Carbotech Germany and global installations.

SUPPORTING POINTS

- The company reports a technology integration and distribution collaboration with Carbotech, Germany, for Carbon Molecular Sieves (CMS) media used in pressure swing adsorption (PSA) gas generation systems.

Geographic Footprint & Sales Network — scored 82/100

> We verified that the company's confirmed physical presence is highly localized, consisting of its corporate office in New Delhi and a single manufacturing plant in Greater Noida. Based on primary research, the company's regional sales and service network can be estimated as comprehensive across major industrial zones, based on its central business district presence in New Delhi and manufacturing facility in Greater Noida.

SUPPORTING POINTS

- The company reports its corporate office is located at Nehru Place in New Delhi and its operational plant or manufacturing facility spans a 25,000 square foot area located in Ecotech-12, Greater Noida, Uttar Pradesh.
- The company reports that its engineered gas generation systems and packages are distributed and installed both within India and internationally across more than 15 countries.

Market Share & Installed Base — scored 77/100

> The assessment rests on the company's verified history as an established provider of industrial gas plants and its self-reported product line of on-site PEM hydrogen systems, though these are typically configured for smaller localized industrial utility use rather than utility-scale green hydrogen projects. Based on primary research, the company's market share and cumulative capacity can be estimated as strong, in line with established industrial gas equipment vendors, based on its 30 years of operational history since 112 and worldwide installations.

SUPPORTING POINTS

- Gastek Engineering Pvt. Ltd. reports that it manufactures custom-engineered on-site hydrogen generators utilizing PEM or alkaline electrolysis technology with capacities ranging from 0.5 to 500 Nm^3/hr, serving industries like steel, pharma, energy, and chemicals across more than 500 total plant installations globally.

Nanosol Energy Private Limited Nanosol Energy ● Evolving Players X 79 Y 73 Overall 76

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 85/100

> We verified that the company has successfully delivered small-scale PEM and AEM test systems to academic and research labs like various IITs. Based on primary research, the company's commercial-scale operational validation can be estimated as early stage or pre-commercial, based on its custom laboratory test benches, single-cell systems, and prototype hydrogen generators delivered to research institutes.

SUPPORTING POINTS

- Nanosol Energy Private Limited has successfully supplied PEM, AEM, and Alkaline electrolyser test systems to various Indian Institutes of Technology (IITs) and R&D institutes.
- The company offers a 3 LPH PEM Hydrogen Generator and an Alkaline electrolyser setup targeted at research and custom equipment fabrication markets.

Electrolyzer System Efficiency & Performance — scored 71/100

> Based on primary research, the company's stack electrochemistry performance can be estimated as tailored for benchtop research optimization, based on its 250 square centimetre AEM and PEM stack hardware designed for experimental validation.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 72/100

> We verified that the company's confirmed capabilities are focused entirely on laboratory-scale and institutional R&D custom hardware fabrication rather than industrial-scale production. Based on primary research, the company's annual manufacturing capacity in mw, factory automation metrics, or commercial capacity expansion plans can be estimated as custom-scale and focused on R&D client requirements, based on its supply of research-grade electrolyzer hardware, fuel cells, and custom test benches to premier academic institutions.

SUPPORTING POINTS

- The company reports designing and developing a 250 cm² area Anion Exchange Membrane (AEM) / Proton Exchange Membrane (PEM) electrolyzer stack hardware.
- The company reports supplying custom-made electrolyzer test systems, single cells, and small-scale stacks—including a 3 LPH PEM hydrogen generator and a 1600W alkaline electrolyzer test system—to institutional research laboratories like CSIR-IICT and various Indian Institutes of Technology (IITs).

Product Portfolio Breadth & Modularity — scored 73/100

> We verified that the company's product capability is confined to bench-scale hardware and custom laboratory-scale R&D test systems rather than commercial or industrial-grade green hydrogen project systems. Based on primary research, the company's production scale capability can be estimated as boutique and custom manufacturing, based on its focus on tailored component fabrication and localized university equipment prototyping. The company's modular architecture expansion can be estimated as limited to laboratory or testing sizes, based on its 3 litres per hour PEM hydrogen generator and custom research scale filtration setups.

SUPPORTING POINTS

- The company reports designing and developing a 250 cm² active area Anion Exchange Membrane (AEM) / Proton Exchange Membrane (PEM) electrolyzer stack hardware.
- The company reports supplying and installing laboratory-scale PEM, AEM, and alkaline electrolyzer test systems (including a 3 liters per hour PEM hydrogen generator and a 1600W alkaline electrolyzer test system) to institutional research clients such as various Indian Institutes of Technology (IITs)

R&D and Innovation Capability — scored 92/100

> We verified that the company possesses technical R&D capabilities focused on small-scale custom stack development and the fabrication of test systems for research institutions. Based on primary research, the company's next-generation development pathway can be estimated as heavily tied to academic innovation hubs, based on its recently signed research and skill development memorandum of understanding with VIT-AP University.

SUPPORTING POINTS

- Nanosol Energy Private Limited has designed, developed, and tested a 250 cm² active area AEM/PEM electrolyser stack hardware for green hydrogen applications.
- The company successfully tested and installed custom AEM and alkaline electrolyser test systems and stacks at research institutions, including the CSIR-Indian Institute of Chemical Technology (CSIR-IICT) and the Indian Institute of Technology (BHU), Varanasi.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's domestic customer contracts or project capacity can be estimated as focused on lab-scale and institutional research applications, based on its successfully supplied PEM, AEM, and Alkaline electrolyser test systems to various IITs and research and development institutes.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 68/100

> The assessment rests on official corporate registry records indicating the company operates on a very small capital foundation. Based on primary research, the company's annual revenue, funding, or financial backing can be estimated as limited and driven primarily by localized equipment sales and consulting fees, based on its startup nature offering basic engineering, process scale-up, and custom fabrication services.

SUPPORTING POINTS

- Ministry of Corporate Affairs records show Nanosol Energy Private Limited was incorporated in December 2020 with an authorized share capital of ₹1,00,000 and a paid-up capital of ₹1,00,000.
- The company operates as a Hyderabad-based green energy startup that develops low-capacity PEM and alkaline electrolysis systems, including a 3 LPH PEM Hydrogen Generator.

Strategic Partnerships & Alliances — scored 77/100

> The rating rests on the company's verified academia-industry research collaboration with VIT-AP University. Based on primary research, the company's joint ventures, technology transfers, or EPC partnerships can be estimated as restricted to custom equipment collaborations and research alliances, based on its test system and stack installation at CSIR-IICT along with custom industrial consultancy.

SUPPORTING POINTS

- The company announced the signing of a Memorandum of Understanding (MoU) with VIT-AP University, Amaravati in 2026 to collaborate in green hydrogen, renewable energy research, skill development, and industry-oriented applications.

Geographic Footprint & Sales Network — scored 75/100

> The rating is based on Nanosol Energy Private Limited operating from a single location in Telangana, serving as its primary engineering and corporate hub, alongside a narrow sales reach focused primarily on supplying R&D laboratory equipment to Indian educational institutes. Based on primary research, the company's sales network or manufacturing expansion can be estimated as concentrated locally within regional technology hubs, based on its core operations being managed directly out of its Hyderabad base.

SUPPORTING POINTS

- Nanosol Energy Private Limited maintains a single physical corporate and operating footprint in India, with its office located at Plot No. 6A, IDA Phase 1, Bandlaguda Village, Patancheru, Telangana.
- The company reports supplying hardware to research institutes across India, including testing systems and fuel cells delivered to IIT Jammu, IIT Varanasi, and Jadavpur University. ·NANOSOL ENERGY PRIVATE LIMITED +2

Market Share & Installed Base — scored 78/100

> The rating rests on the company's active engineering presence in fabricating small-scale PEM and AEM hardware and supplying custom test systems specifically to R&D institutes and universities. Based on primary research, the company's market share or commercial scale deployments can be estimated as niche and early-stage, specialized in test benches and stack hardware, based on its design and development of a 250 square centimetre area AEM and PEM electrolyser stack hardware.

SUPPORTING POINTS

- The company reports designing and developing a 250 cm² area Anion Exchange Membrane (AEM) / Proton Exchange Membrane (PEM) electrolyzer stack hardware intended for green hydrogen applications.
- The company reports successfully supplying several PEM, AEM, and alkaline electrolyzer test systems to various Indian Institutes of Technology (IITs) and R&D institutes for institutional research and performance evaluation.

CLIMAFOUR Labs Pvt. Ltd. CLIMAFOUR ● Evolving Players X 80 Y 67 Overall 74

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 81/100

> The assessment rests on the company's early-stage laboratory development and self-reported performance testing of its Anion Exchange Membrane (AEM) research cell platform. Based on primary research, the company's deployment and operational history can be estimated as early laboratory phase, based on its lab tested performance scaling to kilowatt scale prototypes and working 25 square centimeter cell.

SUPPORTING POINTS

- CLIMAFOUR Labs Pvt. Ltd. has introduced its "CF-AEM Core" research cell line for Anion Exchange Membrane (AEM) water electrolysis, offering cells with 1, 5, and 25 cm² active areas to support catalyst, electrode, and membrane evaluation.
- The company reports that its technology uses non-noble metal catalysts and a binder-free design, achieving lab-tested performance with plans to scale to kilowatt-scale prototypes.

Electrolyzer System Efficiency & Performance — scored 87/100

> The rating rests on the company's early-stage development of Anion Exchange Membrane (AEM) electrolyzer cells and hardware for laboratory research and validation. Based on primary research, the company's commercial efficiency and degradation can be estimated as material level experimental validation, based on its research cell family for screening catalysts and device level validation.

SUPPORTING POINTS

- The company reports progressing from material-level testing to 100 cm² coated electrodes and demonstrating a working 25 cm² AEM electrolyzer cell at 25 A (1.0 A cm^{-2}).

Manufacturing Capacity & Scalability — scored 67/100

> We verified that the company operates as an early-stage research startup whose capabilities are limited to laboratory-scale hardware and working on a ~2 kW multi-cell prototype. Based on primary research, the company has not disclosed any commercial-scale manufacturing facilities, automated factories, annual production capacity in MW, or plans for capacity expansion.

SUPPORTING POINTS

- The company reports it has progressed from material-level testing to 100 cm² coated electrodes and a working 25 cm² Anion Exchange Membrane (AEM) electrolyzer cell demonstrated at 25 A, with a next engineering milestone focused on building a ~2 kW multi-cell prototype.

Product Portfolio Breadth & Modularity — scored 73/100

> The rating is based on the company's confirmed commercial portfolio being restricted to laboratory-scale and R&D research hardware spanning cell sizes up to 25 cm². Based on primary research, the company's commercial electrolyzer product range can be estimated as foundational research scale hardware offerings, based on its launch of the commercial CF-AEM Core research cell family.

SUPPORTING POINTS

- The company reports offering the CLIMAFOUR Research Electrolyzer Platform, which consists of standardized 1 cm², 5 cm², and 25 cm² Anion Exchange Membrane (AEM) research electrolyzer hardware designed for catalyst screening, membrane-electrode assembly (MEA) development, and device-level testing.
- The company reports that its current technology scale has progressed to 100 cm² coated electrodes and an operational 25 cm² AEM single cell operating at 25 A, with a ~2 kW multi-cell prototype established as its next engineering milestone.

R&D and Innovation Capability — scored 90/100

> The rating rests on the company's early-stage development and commercial introduction of research cell kits designed to optimize and test anion exchange membrane (AEM) components. Based on primary research, the company has not disclosed audited R&D expenditures, its specific domestic or international patent portfolio, or formal, institutionalized corporate-academic collaborative research pipelines.

SUPPORTING POINTS

- CLIMAFOUR Labs Pvt. Ltd. reports developing the "CF-AEM Core" research cell family for AEM water electrolysis, featuring active areas of 1, 5, and 25 cm² to support early material screening, catalysts, electrodes, and membrane evaluation.
- The company reports utilizing non-noble metal catalysts and a binder-free design free of PFAS and Nafion, achieving lab-tested performance that is transitioning to kilowatt-scale prototypes.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 65/100

> Ltd. Based on primary research, the company's client identities and project capacities can be estimated as limited to laboratory researchers, based on its standardized 31 square centimeter electrolyzer cells and focus on catalyst screening.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 68/100

> The rating rests on the company's status as an early-stage startup developing Anion Exchange Membrane (AEM) research hardware platforms, with its financial backing limited to minimal initial capital raised from its founders. Based on primary research, the company's annual commercial revenues and institutional funding can be estimated as negligible and early stage, based on its 11,800 dollars raised from its founders.

SUPPORTING POINTS

- CLIMAFOUR Labs Pvt. Ltd. raised $11,800 in funding from internal founders and individual backers.

Strategic Partnerships & Alliances — scored 69/100

> Based on primary research, the company's commercial alliances can be estimated as restricted to academic collaborations, based on its global search for researchers and technologists to evaluate material level testing.

SUPPORTING POINTS

- The company has announced that research enquiries, academic collaborations, and commercial R&D partnerships are open for its AEM electrolyzer component and hardware development.

Geographic Footprint & Sales Network — scored 65/100

> We verified that the company's confirmed physical presence is limited to its startup laboratory and registered facility in Jaipur, Rajasthan, with no established regional sales networks or multi-city distribution hubs in place yet. Based on primary research, the company's regional footprint and manufacturing infrastructure can be estimated as localized to a single laboratory space, based on its jaipur base and focus on benchtop prototyping.

SUPPORTING POINTS

- The company maintains its corporate headquarters and registered office at Plot No. 77, East Part 21, South Niwaru Road, Jhotwara, Jaipur, Rajasthan, India.
- The company reports its next engineering milestone is scaling to a 2 kW multi-cell prototype, with commercial research cells available for order via inquiry.

Market Share & Installed Base — scored 69/100

> We verified that the company operates as an early-stage startup focused on developing lab-scale Anion Exchange Membrane (AEM) research electrolyzer platforms (1, 5, and 25 cm² cell sizes) and working toward a 2 kW multi-cell prototype, but it has not deployed commercial-scale systems. Based on primary research, the company's market share and cumulative operational capacity sold can be estimated as virtually nonexistent in commercial setups, based on its 0 commercial megawatts deployed and upcoming milestone of a 2 kilowatt multi cell prototype.

SUPPORTING POINTS

- Not available

Alphios Finergy Private Limited

Alphios Finergy · Evolving Players · X 72 · Y 68 · Overall 70

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 71/100

> We verified that the company's stack technology capability resides at an early concept or development stage, as it lacks publicly confirmed commercial scale operational hours or third-party performance validations. Based on primary research, the company's operational hours and deployments can be estimated as pre-commercial, focused on pilot validation and development cohorts, based on its partnership with Link Innovations and acceptance into the Baker Hughes energy innovation cohort.

SUPPORTING POINTS

- The company reports aiming to develop Anion Exchange Membrane (AEM) electrolyzer technology to reduce stack component and manufacturing costs by 60%.

Electrolyzer System Efficiency & Performance — scored 76/100

> We verified that Alphios Finergy Private Limited's product capabilities rest on an early-stage partnership with IIT Madras to develop a wastewater-compatible electrolyzer prototype. Based on primary research, the company has not disclosed real operating condition metrics such as electrical efficiency (kWh/kg H2), current density, or long-term system degradation rates.

SUPPORTING POINTS

- The company reports developing a bi-membrane and a bifunctional catalyst flow cell technology in collaboration with IIT Madras, designed to utilize highly corrosive industrial wastewater feed without needing elevated temperatures.

Manufacturing Capacity & Scalability — scored 65/100

> Based on primary research, the company's manufacturing capacity and expansion plans can be estimated as early-stage and building toward localized deep-tech assembly, based on its incubation at the Finblue Software Technology Parks of India hub and onboarding at the iTNT deep-tech pavilion.

SUPPORTING POINTS

- Not available

Product Portfolio Breadth & Modularity — scored 68/100

> We verified that the company has conceptual or early-stage development claims for an AEM and integrated thermo-electrolysis platform. Based on primary research, the company's commercialized product portfolio can be estimated as expanding toward utility scale, based on its design of a 10 megawatt green hydrogen hub powered by renewable virtual power plants.

SUPPORTING POINTS

- The company reports developing an Anion Exchange Membrane (AEM) electrolysis-based electrolyzer technology that aims to combine the advantages of alkaline and PEM systems using earth-abundant materials.
- The company claims to be building an electro-thermal H2-CO2 electrolysis system or a physics-informed neural network (PINN) inferenced hydrogen-syngas reactor designed to lower capital expenditures.

R&D and Innovation Capability — scored 81/100

> The rating is based on the company's early-stage development of its proprietary AEM electrolysis stack concept designed to combine alkaline and PEM advantages. Based on primary research, the company's internal research and development capability can be estimated as robust and industry validated, based on its selection by the Baker Hughes energy innovation cohort and partnerships with STPI FinBlue and iTNT Hub.

SUPPORTING POINTS

- The company reports developing an innovative Anion Exchange Membrane (AEM) electrolysis technology that combines the features of alkaline and PEM systems, intending to lower manufacturing costs by up to 60%.

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> We verified that Alphios Finergy Private Limited is an early-stage deep-tech startup developing integrated thermo-electrochemical and chemical looping pathways for hydrogen production. Based on primary research, the company's signed domestic customer contracts can be estimated as limited to a few projects, based on its incorporation as an early stage venture and its focus on clean tech green hydrogen production via anion exchange membrane electrolysis technology.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 65/100

> Based on primary research, the company's annual revenue or total funding can be estimated as growing steadily from a low base, based on its generation of less than 10 crore rupees in revenue for the financial year ending on march 30 first 2,025.

SUPPORTING POINTS

- Not available

Strategic Partnerships & Alliances — scored 67/100

> Based on primary research, the company's technology, epc, or conglomerate collaborations can be estimated as emerging through targeted incubator programs, based on its technology benefits of reducing manufacturing costs by 60% and its incubation at finblue software technology parks of india chennai.

SUPPORTING POINTS

- Not available

Geographic Footprint & Sales Network — scored 71/100

> We verified that the company's confirmed physical presence is restricted to its single registered and incubation site in Chennai, Tamil Nadu. Based on primary research, the company's additional offices, facilities, or distribution agreements can be estimated as centralized at its primary headquarters, based on its registered office located at plot number 62 sulochana nagar kundrathur main road moulivakkam chennai.

SUPPORTING POINTS

- Alphios Finergy Private Limited is an early-stage clean-tech startup incorporated in January 2023 with its registered office located at P.No. 62, Sulochana Nagar, Kundrathur Main Road, Moulivakkam, Chennai, Tamil Nadu.
- The company is developing integrated electrolysis solutions and has been incubated at Finblue - Software Technology Parks of India (STPI) in Chennai, but it maintains no other confirmed offices, manufacturing sites, or active distribution partnerships across other regions of India.

Market Share & Installed Base — scored 69/100

> Based on primary research, the company's market share or cumulative installed capacity can be estimated as minor but pioneering in the specialized alternative technology segment, based on its deployment of anion exchange membrane technology aimed at lowering green hydrogen costs from 5.5 dollars per kilogram to 2 dollars per kilogram.

SUPPORTING POINTS

- The company reports it is building an integrated thermo-electrochemical platform to produce green hydrogen and syngas from industrial wastewater and carbon dioxide, and in February 2026, it stated it was in discussions with potential pilot partners in the oil and gas and steel industries.
- Not available ·Alphios Finergy +1

Hydrovolt Energy Ptx Private Limited

Hydrovolt Energy · Evolving Players · X 72 · Y 67 · Overall 70

X — PRODUCT CAPABILITY

PEM/AEM Stack Technology Readiness — scored 65/100

> Based on primary research, the company's technical specifications, validation data, or active projects can be estimated as customized to project requirements, based on its design, construction, and deployment of green hydrogen solutions for advanced industrial applications.

SUPPORTING POINTS

- Not available

Electrolyzer System Efficiency & Performance — scored 65/100

> Based on primary research, the company's electrical efficiency, current density, or degradation rates can be estimated as optimized for advanced industrial deployment, based on its tailor-made alkaline, PEM, and AEM electrolyzer solutions built to enhance operational efficiency.

SUPPORTING POINTS

- Not available

Manufacturing Capacity & Scalability — scored 72/100

> The rating rests on the company's verified role as an early-stage startup developer offering customized, end-to-end integration and supply of multi-technology electrolysers. Based on primary research, the company's manufacturing facilities and capacity expansions can be estimated as managed through its dedicated engineering and tech setup, based on its corporate office in Noida and registered office in Ghaziabad providing end-to-end supply.

SUPPORTING POINTS

- The company reports supplying Alkaline Water Electrolyser (AWE), PEM, and AEM electrolysers tailor-made to specific specifications and green hydrogen project requirements.
- The company has deployed a fuel cell for IIT Bombay and is in discussions with clients for green hydrogen projects.

Product Portfolio Breadth & Modularity — scored 86/100

> Our assessment rests on the company's self-reported capabilities to supply multiple types of electrolyzer solutions and integrated lab systems. Based on primary research, the company's stack capacities and modular scaling options can be estimated as tailor-made to specific hydrogen requirements, based on its diverse configurations across alkaline, PEM, and AEM technologies.

SUPPORTING POINTS

- The company reports that it delivers Alkaline, Proton Exchange Membrane (PEM), and Anion Exchange Membrane (AEM) electrolyzer solutions customized to project specifications and requirements.
- The company reports offering integrated green hydrogen lab solutions and on-site ultra-high purity hydrogen generators.

R&D and Innovation Capability — scored 71/100

> The rating rests on the absence of publicly available information regarding the company's research infrastructure, proprietary technical pipelines, or patent portfolio specifically for PEM and AEM electrolyzer systems. Based on primary research, the company's research expenditure, patent filings, or institutional collaborations can be estimated as driven by business strategy and technological acumen, based on its specialist team combining deep expertise across renewable energy, finance, and business strategy.

SUPPORTING POINTS

- Not available

Y — BUSINESS CAPABILITY

Domestic Customer Contracts & Order Book — scored 66/100

> Based on primary research, the company's domestic order book can be estimated as early stage and focused on local public sector contracts, based on its rate contract won with Bharat Heavy Electricals Limited and public bids with the Ministry of Earth Sciences.

SUPPORTING POINTS

- Not available

Revenue & Funding Strength — scored 65/100

> Based on primary research, the company's financial performance can be estimated as emerging and self-sustained, based on its unfunded startup operational baseline established by founders Akash Goel and Karan Goyal.

SUPPORTING POINTS

- Not available

Strategic Partnerships & Alliances — scored 67/100

> Based on primary research, the company's project collaborations can be estimated as highly specialized and client-specific, based on its customized balance of plant, consultancy, and end-to-end engineering service models.

SUPPORTING POINTS

- Not available

Geographic Footprint & Sales Network — scored 72/100

> The rating rests on the company's confirmed presence via its corporate office in Noida and registered office in Ghaziabad. Based on primary research, the company's additional offices, dedicated manufacturing facilities, sales/service locations, or distribution agreements can be estimated as limited to a few markets, based on its corporate office in Noida and registered office in Ghaziabad.

SUPPORTING POINTS

- The company maintains its corporate office at Candor Techspace Tower 5, 14th Floor, Sector 62, Noida, Uttar Pradesh, and its registered office at A-4, Lohia Nagar, Ghaziabad, Uttar Pradesh.

Market Share & Installed Base — scored 65/100

> Based on primary research, the company's installed base can be estimated as nascent and growing, based on its end-to-end tailored system supply strategy for custom alkaline, proton exchange membrane, and anion exchange membrane electrolyzers.

SUPPORTING POINTS

- Not available

Parameter Definitions

X — Product Capability

1. **PEM/AEM Stack Technology Readiness**
   The maturity and proven reliability of the company's PEM or AEM stack technology, evidenced by operational hours, third-party certifications, and successful deployment in pilot or commercial projects.
2. **Electrolyzer System Efficiency & Performance**
   The electrical efficiency (kWh/kg H2) and current density of the electrolyzer system under real operating conditions, with higher efficiency and lower degradation rates indicating superior product strength.
3. **Manufacturing Capacity & Scalability**
   The company's ability to mass-produce PEM/AEM electrolyzers, including factory automation, annual production capacity in MW, and plans for capacity expansion to meet large project demands.
4. **Product Portfolio Breadth & Modularity**
   The range of electrolyzer products offered (e.g., stack-only vs. integrated systems, different sizes) and their modularity for easy scaling, which determines suitability for diverse project sizes.
5. **R&D and Innovation Capability**
   The company's investment in research and development, patent portfolio, collaboration with research institutions, and pipeline of next-generation improvements in PEM/AEM technology.

Y — Business Capability

1. **Domestic Customer Contracts & Order Book**
   The value and number of signed contracts or letters of intent from Indian green hydrogen project developers, including names of customers and project capacities (in MW).
2. **Revenue & Funding Strength**
   The company's annual revenue from electrolyzer sales, recent funding rounds, and financial backing from investors or government schemes, indicating commercial viability and growth potential.
3. **Strategic Partnerships & Alliances**
   Collaborations with technology providers, EPC firms, or industrial conglomerates that enhance market access, technology transfer, or project execution, evidenced by named partners and joint ventures.
4. **Geographic Footprint & Sales Network**
   The company's presence across India, including offices, manufacturing sites, and sales/service locations, as well as distribution agreements that enable reach to customers in different regions.
5. **Market Share & Installed Base**
   The company's share of the Indian PEM/AEM electrolyzer market by capacity sold or installed, and the cumulative installed base of electrolyzers in operational green hydrogen projects.

Research Methodology

This quadrant was compiled independently through structured secondary research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1 Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

→

2 Secondary and Primary Research

Secondary research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

→

3 Screening

Each candidate was screened against the market's inclusion criteria and required commercial role. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicates consolidated. Companies whose main business is an adjacent segment are listed but not placed on the chart.

→

4 Evidence Compilation

Every screened company was assessed on all ten parameters; for chart companies, evidence was recorded parameter by parameter. Claims resting only on a company's own materials are reported as company-reported, partner technology is credited to the partnership, and employee-review sites are not used as customer evidence.

→

5 Scoring

Each parameter is rated 0–100 against its written definition, based on the evidence available. Ratings are analyst assessments under this methodology, not independently audited figures. X and Y are the simple average of the five parameter ratings on each axis; overall is the average of X and Y.

→

6 Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.